title: "Embroidermodder 2.0.0-alpha" subtitle: "Reference Manual" date: "2023-10-02" toc: true math: true ---

Embroidermodder 2.0.0-Alpha

The Embroidermodder Team <embroidermodder@gmail.com> :description: Reference Manual for the technical details of all of the Embroidermodder Projects :url-repo: https://github.com/embroidermodder/www.libembroidery.org

Introduction

Test these:

```
sudo apt install latexml texlive-latex-base imagemagick info2man

# For our command line tools:
makeinfo embroider.texi -o embroider.info
info2man embroider.info > embroider.1
texi2pdf embroider.texi
# Or groff macro package for example ms.
# These may be housed in libembroidery since they're to be shipped as part of
# the embroider tarball.

# For online documentation:
pandoc embroidermodder_refman.tex -f latex -t html -s -o emb_refman.html
--bibliography embroidermodder.bib
# Or latexml/latexmlpost
```

Command Language

Printer Command Language (PCL), see \citet{packard1992pcl}.

HP-GL/2 Vector Graphics \index{HP-GL/2} described in \citet{packard1992pcl}. Has commands like: "PU" Pen Up, "PR" Plot Relative, "EP" edge polygon.

So commands read like this:

```
PA40,10;
```

command argument seperator(`,`) argument terminator(`;`)

Constructing new commands from old ones in the command language is less natural in the HP-GL/2 language, but a similar layer for us is the tajima DST format %\citet{4} for existing printers and CNC commands for direct control... where'd we'd use G-Code %\citet{7} and Linux CNC \citet{6}.

Could we \texttt{setpagedevice} to a printer in some cases and a similar CUPS service for embroidery machines in others?

All systems are supported by ghostscript, when you account for homebrew %\citet{9}:

```
brew update
brew upgrade
brew install ghostscript
brew cleanup
```

Vector graphic logos don't require the SVG Qt library.

\section{Man Pages}

We maintain a traditional manpage for \texttt{embroider} using the basic macros.

\section{Arduino}

\begin{verbatim} apt-get install avr-libc gcc-avr uisp avrdude \end{verbatim}

\section{thingy}

\chapter{Libembroidery}

(Under construction, please wait for v1.0 release.)

Libembroidery is a low-level library for reading, writing, and altering digital embroidery files in C. It is part of the Embroidermodder Project for open source machine embroidery.

Libembroidery is the underlying library that is used by Embroidermodder 2 and is developed by The Embroidermodder Team \ref{the-embroidermodder-team}. A full list of contributors to the project is maintained in \url{https://github.com/Embroidermodder/embroidermodder} the Embroidermodder 2 github in the file \texttt{CREDITS.md}. It handles over 45 different embroidery specific formats as well as several non-embroidery specific vector formats.

It also includes a CLI called \texttt{embroider} that allows for better automation of changes to embroidery files and will be more up-to date than the Embroidermodder 2 GUI.

Documentation

Libembroidery is documented as part of this reference manual. If you need libembroidery for any non-trivial usage or want to contribute to the library we advise you read the appropriate design sections of the manual first. Copies of this manual will be shipped with the packaged version of libembroidery, but to build it we use the Doxyfile in \url{https://github.com/Embroidermodder/embroidermodder} the Embroidermodder git repository.

For more basic usage, \texttt{embroider} should have some in-built help starting with:

```
$ embroider --help
```

License

Libembroidery is distributed under the permissive zlib licence, see the LICENCE file.

Demos

We're currently trying out some fill techniques which will be demonstrated here and in the script \texttt{qa_test.sh}.

\begin{center} \includegraphics[width=0.5\textwidth]{images/logo.png} \end{center}

Converts to:

\begin{center} \includegraphics[width=0.5\textwidth]{images/crossstitch_logo.png} \end{center}

Build

libembroidery and EmbroiderModder 2 use CMake builds so if you are building the project to use as a library we recommend you run:

```
git clone https://github.com/Embroidermodder/libembroidery
cd libembroidery
cmake .
cmake --build .
cmake --install .
```

This builds both the static and shared versions of the library as well as the command line program \texttt{embroider}.

\citet{packard1992pcl}

\citet{linuxcncsrc}

\citet{linuxcnc}

\citet{adobe1990postscript}

\citet{postscript1999postscript}

\citet{eduTechDST}

\citet{cups}

\citet{millOperatorsManual}

\citet{oberg1914machinery}

\citet{dxf_reference}

\citet{embroidermodder_source_code}

\citet{libembroidery_source_code}

\citet{acatina}

\citet{kde_tajima}

\citet{wotsit_archive}

\citet{wotsit_siterip}

\citet{fineemb_dst}

\citet{edutechwiki_dst}

\chapter{Graphical User Interface for PC} \ref{GUI}

Overview

\begin{center} (*UNDER MAJOR RESTRUCTURING, PLEASE WAIT FOR VERSION 2*)

https://www.libembroidery.org \end{center}

Embroidermodder is a free machine embroidery application. The newest version, Embroidermodder 2 can:

- edit and create embroidery designs
- estimate the amount of thread and machine time needed to stitch a design
- convert embroidery files to a variety of formats
- upscale or downscale designs
- run on Windows, Mac and Linux

Embroidermodder 2 is very much a work in progress since we're doing a ground up rewrite to an interface in C using the GUI toolkit SDL2. The reasoning for this is detailed in the issues tab.

For a more in-depth look at what we are developing read our website (\url{https://www.libembroidery.org}) which includes these docs as well as the up-to date printer-friendly versions. These discuss recent changes, plans and has user and developer guides for all the Embroidermodder projects.

To see what we're focussing on right now, see the Open Collective News (\url{https://opencollective.com/embroidermodder}).

This current printer-friendly version is here (\url{https://www.libembroidery.org/EM2.0.0-alpha_refman_a4.pdf}).

License

The source code is under the terms of the zlib license: see \texttt{LICENSE.md} in the source code directory.

Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.3 or any later version published by the Free Software Foundation; with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts.

A copy of the license is included in Section~\ref{GNU-free-documentation-license}.

The Embroidermodder Project and Team

The \emph{Embroidermodder 2} project is a collection of small software utilities for manipulating, converting and creating embroidery files in all major embroidery machine formats. The program \emph{Embroidermodder 2} itself is a larger graphical user interface (GUI) which is at the heart of the project.

The tools and associated documents are:

- The website \url{https://www.libembroidery.org}
- This reference manual covering the development of all these projects with the current version available here: \url{https://www.libembroidery.org/embroidermodder_2.0_manual.pdf}
- The GUI \emph{Embroidermodder 2} covered in Chapter~\ref{GUI}.
- The core library of low-level functions: \texttt{libembroidery}, covered in Chapter~\ref{libembroidery}
- The CLI \texttt{embroider}, which is part of libembroidery
- Mobile embroidery format viewers and tools convered in Chapter~\ref{Mobile}.
- Specs for an open source hardware embroidery machine extension called the Portable Embroidery Tool (PET) which is also part of libembroidery. See Chapter~\ref{PET}.

The website, this manual and some scripts to construct the distribution are maintained in \citet{embroidermodder_website}.

They all tools to make the standard user experience of working with an embroidery machine better without expensive software which is locked to specific manufacturers and formats. But ultimately we hope that the core \emph{Embroidermodder 2} is a practical, ever-present tool in larger workshops, small cottage industry workshops and personal hobbyist's bedrooms.

Embroidermodder 2 is licensed under the zlib license and we aim to keep all of our tools open source and free of charge. If you would like to support the project check out our Open Collective (\url{https://opencollective.com/embroidermodder}) group. If you would like to help, please join us on GitHub. This document is written as developer training as well helping new users (see the last sections) so this is the place to learn how to start changing the code.

The Embroidermodder Team is the collection of people who've submitted patches, artwork and

documentation to our three projects. The team was established by Jonathan Greig and Josh Varga. The full list is actively maintained below.

Core Development Team

Embroidermodder 2:

- Jonathan Greig (https://github.com/redteam316)
- Josh Varga (https://github.com/JoshVarga)
- Robin Swift (https://github.com/robin-swift)

Embroidermodder 1:

- Josh Varga (https://github.com/JoshVarga)
- Mark Pontius (http://sourceforge.net/u/mpontius/profile)

Credits for Embroidermodder 2, libembroidery and all other related code

If you have contributed and wish to be added to this list, alter the README on Embroidermodder github page (https://github.com/Embroidermodder/Embroidermodder) and we'll copy it to the libembroidery source code since that is credited to "The Embroidermodder Team".

Embroidermodder 1

The Embroidermodder Team is also inspired by the original Embroidermodder that was built by Mark Pontius and the same Josh Varga on SourceForge which unfortunately appears to have died from linkrot. We may create a distribution on here to be the official "legacy" Embroidermodder code but likely in a seperate repository because it's GNU GPL v3 and this code is written to be zlib (that is, permissive licensed) all the way down.

One reason why this is useful is that the rewrite by Jonathan Greig, John Varga and Robin Swift for Embroidermodder 2 should have no regressions: no features present in v1 should be missing in v2.

Features

Embroidermodder 2 has many advanced features that enable you to create awesome designs quicker, tweak existing designs to perfection, and can be fully customized to fit your workflow.

A summary of these features:

\begin{itemize} \item Cross Platform \item Realistic rendering \item Various grid types and auto-adjusting rulers \item Many measurement tools \item Add text to any design \item Supports many formats \item Batch Conversion \item Scripting API \end{itemize}

Cross Platform}

If you use multiple operating systems, it’s important to choose software that works on all of them.

Embroidermodder 2 runs on Windows, Linux and Mac OS X. Let’s not forget the Raspberry Pi (\url{http://www.raspberrypi.org}).

\begin{figure} \begin{center} \includegraphics[width=0.6\textwidth]{images/features-platforms-1.png} \end{center} \caption{Running on different platforms.} \end{figure}

Realistic Rendering}

It is important to be able to visualize what a design will look like when stitched and our pseudo ``3D'' realistic rendering helps achieve this (see Figure~\ref{real-render}).

\begin{figure} \begin{center} \includegraphics[width=0.6\textwidth]{images/features-realrender-1.png} \includegraphics[width=0.6\textwidth]{images/features-realrender-2.png} \includegraphics[width=0.6\textwidth]{images/features-realrender-3.png} \end{center} \caption{Real render examples.} \label{fig:real-render} \end{figure}

Various grid types and auto-adjusting rulers}

Making use of the automatically adjusting ruler in conjunction with the grid will ensure your design is properly sized and fits within your embroidery hoop area.

Use rectangular, circular or isometric grids to construct your masterpiece!

Multiple grids and rulers in action Figure~\ref{fig:grid-ruler}.

\begin{figure} \begin{center} \includegraphics[width=0.6\textwidth]{images/features-grid-ruler-1.png} \end{center} \caption{Grid and ruler examples} \label{fig:grid-ruler} \end{figure}

Many measurement tools}

Taking measurements is a critical part of creating great designs. Whether you are designing mission critical embroidered space suits for NASA or some other far out design for your next meet-up, you will have precise measurement tools at your command to make it happen. You can locate individual points or find distances between any 2 points anywhere in the design!

Take quick and accurate measurements:

! features measure 1 (images/features-measure-1.png)

Add text to any design}

Need to make company apparel for all of your employees with individual names on them? No sweat. Just simply add text to your existing design or create one from scratch, quickly and easily. Didn’t get it the right size or made a typo? No problem. Just select the text and update it with the property editor.

Add text and adjust its properties quickly:

! features text 1 (images/features-text-1.png)

Supports many formats

Embroidery machines all accept different formats. There are so many formats available that it can sometimes be confusing whether a design will work with your machine.

Embroidermodder 2 supports a wide variety of embroidery formats as well as several vector formats, such as SVG and DXF. This allows you to worry less about which designs you can use.

Batch Conversion

Need to send a client several different formats? Just use libembroidery-convert, our command line utility which supports batch file conversion.

There are a multitude of formats to choose from:

! features formats (images/features-formats-1.png)

Scripting API

If you've got programming skills and there is a feature that isn't currently available that you absolutely cannot live without, you have the capability to create your own custom commands for Embroidermodder 2. We provide an QtScript API which exposes various application functionality so that it is possible to extend the application without requiring a new release. If you have created a command that you think is worth including in the next release, just contact us (contact.html) and we will review it for functionality, bugs, and finally inclusion.

An Embroidermodder 2 command excerpt:

! scripting screenshot (images/features-scripting-1.png)

Build and Install}

Assuming you already have the SDL2 libraries you can proceed to using the fast build, which assumes you want to build and test locally.

The fast build should be:

\begin{lstlisting} bash build.sh \end{lstlisting}

or, on Windows:

\begin{lstlisting} .\build.bat \end{lstlisting}

Then run using the \texttt{run.bat} or \texttt{run.sh} scripts in the build/ directory.

Otherwise, follow the instructions below.

If you plan to install the dev version to your system (we recommend you wait for the official installers and beta release first) then use the CMake build instead.

Install on Desktop}

We recommend that if you want to install the development version you use the CMake build. Like this:

\begin{lstlisting} git submodule init git submodule update

mkdir build cd build cmake .. cmake --build . sudo cmake --install . \end{lstlisting}

These lines are written into the file:

\begin{lstlisting} ./build_install.sh \end{lstlisting}

On Windows use the next section.

History}

Embroidermodder 1 was started by Mark Pontius in 2004 while staying up all night with his son in his first couple months. When Mark returned to his day job, he lacked the time to continue the project. Mark made the decision to focus on his family and work, and in 2005, Mark gave full control of the project to Josh Varga so that Embroidermodder could continue its growth.

Embroidermodder 2 was conceived in mid 2011 when Jonathan Greig and Josh Varga discussed the possibility of making a cross-platform version. It is currently in active development and will run on GNU/Linux, Mac OS X, Microsoft Windows and Raspberry Pi.

All Embroidermodder downloads (downloads.html) are hosted on SourceForge.

The source code for Embroidermodder 1 (\url{http://sourceforge.net/p/embroidermodder/code/HEAD/tree/embroidermodder1}) has always been hosted on Sourceforge.

The source code for Embroidermodder 2 (\url{https://github.com/Embroidermodder/Embroidermodder}) was moved to GitHub on July 18, 2013.

The website for Embroidermodder (\url{https://github.com/Embroidermodder/www.libembroidery.org}) was moved to GitHub on September 9, 2013.

Contact us

For general questions email: embroidermodder at gmail.com (mailto:embroidermodder@gmail.com)

To request a new feature open an issue on the main Embroidermodder GitHub repository

(https://github.com/Embroidermodder/Embroidermodder/issues). We'll move it to the correct repository.

Downloads

Alpha Build

This is a highly experimental build: we recommend users wait for the beta release when the basic features are functional.

Visit our GitHub Releases page (https://github.com/Embroidermodder/Embroidermodder/releases) for the current build. Unfortunately, earlier builds went down with the Sourceforge page we hosted them on.

•

Embroidermodder 2 is very much a work in progress since we're doing a ground up rewrite to an interface in Python using the GUI toolkit Tk. The reasoning for this is detailed in the issues tab.

For a more in-depth look at what we are developing read the developer notes (link to dev notes section). This discusses recent changes in a less formal way than a changelog (since this software is in development) and covers what we are about to try.

Documentation

The documentation is in the form of the website (included in the \texttt{docs/} directory) and the printed docs in this file.

Development

If you wish to develop with us you can chat via the contact email on the website (\url{https://www.libembroidery.org}) or in the issues tab on the github page (\url{https://github.com/Embroidermodder/Embroidermodder/issues}). People have been polite and friendly in these conversations and I (Robin) have really enjoyed them. If we do have any arguments please note we have a Code of Conduct (\texttt{CODE_OF_CONDUCT.md}) so there is a consistent policy to enforce when dealing with these arguments.

The first thing you should try is building from source using the build advice(link to build) above. Then read some of the development notes (link to dev notes.md) to get the general layout of the source code and what we are currently planning.

Testing

To find unfixed errors run the tests by launching from the command line with:

\begin{lstlisting} $ embroidermodder --test \end{lstlisting}

then dig through the output. It's currently not worth reporting the errors, since there are so many but if you can fix anything reported here you can submit a PR.

Code Optimisations and Simplifications

Geometry

The geometry is stored, processed and altered via libembroidery. See the Python specific part of the documentation for libembroidery for this. What the code in Embroidermodder does is make the GUI widgets to change and view this information graphically.

For example if we create a circle with radius 10mm and center at \texttt{(20mm, 30mm)} then fill it with stitches the commands would be

\begin{lstlisting} from libembroidery import Pattern, Circle, Vector, satin circle = Circle(Vector(20, 30), 10) pattern = Pattern() pattern.add_circle(circle, fill=satin) pattern.to_stitches() \end{lstlisting}

but the user would do this through a series of GUI actions:

1. Create new file
2. Click add circle
3. Use the Settings dialog to alter the radius and center
4. Use the fill tool on circle
5. Select satin from the drop down menu

So EM2 does the job of bridging that gap.

Postscript Support

In order to safely support user contributed/shared data that can define, for example, double to double functions we need a consistent processor for these descriptions.

Embroidermodder backends to the postscript interpreter included in libembroidery to accomplish this.

For example the string: `5 2 t mul add` is equivalent to the expression $2*t + 5$.

The benefit of not allowing this to simply be a Python expression is that it is safe against malicious use, or accidental misuse. The program can identify whether the output is of the appropriate form and give finitely many calculations before declaring the function to have run too long (stopping equations that hang).

To see examples of this see the \texttt{assets/shapes/*.ps} files.

SVG Icons

To make the images easier to alter and restyle we could switch to svg icons. There's some code in the git history to help with this.

The Actions System

In order to simplify the development of a GUI that is flexible and easy to understand to new developers we have a custom action system that all user actions will go via an \texttt{actuator} that takes a string argument. By using a string argument the undo history is just an array of strings.

The C \texttt{action_hash_data} struct will contain: the icon used, the labels for the menus and tooltips and the function pointer for that action. There will be an accompanying argument for this function call, currently being drafted as \texttt{action_call}. So when the user makes a function call it should contain information like the mouse position, whether special key is pressed etc.

Accessibility

Software can be more or less friendly to people with dylexia, partial sightedness, reduced mobility and those who don't speak English. Embroidermodder 2 has, in its design, the following features to help:

- icons for everything to reduce the amount of reading required
- the system font is configurable: if you have a dyslexia-friendly font you can load it
- the interface rescales to help with partial-sightedness
- the system language is configurable, unfortunately the docs will only be in English but we can try to supply lots of images of the interface to make it easier to understand as a second language
- buttons are remappable: XBox controllers are known for being good for people with reduced mobility so remapping the buttons to whatever setup you have should help

Note that most of these features will be released with version 2.1, which is planned for around early 2023.

Sample Files

Various sample embroidery design files can be found in the `embroidermodder2/samples` folder.

Shortcuts

A shortcut can be made up of zero or more modifier keys and at least one non-modifier key pressed at once.

To make this list quickly assessable, we can produce a list of hashes which are simply the flags ORed together.

The shortcuts are stored in the csv file \texttt{shortcuts.csv} as a 5-column table with the first 4 columns describing the key combination. This is loaded into the shortcuts \texttt{TABLE}. Each tick the program checks the input state for this combination by first translating the key names into indices for the key state, then checking for whether all of them are set to true.

Removed Elements

So I've had a few pieces of web infrastructure fail me recently and I think it's worth noting. An

issue that affects us is an issue that can effect people who use our software.

Qt and dependencies

Downloading and installing Qt has been a pain for some users (46Gb on possibly slow connections).

I'm switching to FreeGLUT 3 (which is a whole other conversation) which means we can ship it with the source code package meaning only a basic build environment is necessary to build it.

Social Platform

Github is giving me a server offline (500) error and is still giving a bad ping.

So… all the issues and project boards etc. being on Github is all well and good assuming that we have our own copies. But we don't if Github goes down or some other major player takes over the space and we have to move (again, since this started on SourceForge).

This file is a backup for that which is why I'm repeating myself between them.

OpenGL

OpenGL rendering within the application. This will allow for Realistic Visualization - Bump Mapping/OpenGL/Gradients?

This should backend to a C renderer or something.

Configuration Data Ideas

Embroidermodder should boot from the command line regardless of whether it is or is not installed (this helps with testing and running on machines without root). Therefore, it can create an initiation file but it won't rely on its existence to boot: \texttt{~/.embroidermodder/config.json}.

\begin{itemize} \item Switch colors to be stored as 6 digit hexcodes with a \texttt{\#}. \item We've got close to a hand implemented ini read/write setup in \texttt{settings.py}. \end{itemize}

Distribution}

\index{distribution}

When we release the new pip wheel we should also package:

\begin{itemize} \item \texttt{.tar.gz} and \texttt{.zip} source archive. \item Debian package \item RPM package \end{itemize}

Only do this once per minor version number.

\begin{itemize} \item todo Screenshot a working draft to demonstrate. \end{itemize}

Perennial Jobs}

\begin{itemize} \item Check for memory leaks \item Clear compiler warnings on \texttt{-Wall -ansi -pedantic} for C. \item Write new tests for new code. \item Get Embroidermodder onto the current version of libembroidery. \item PEP7 compliance. \item Better documentation with more photos/screencaps. \end{itemize}

Full Test Suite}

\index{testing}

(This needs a hook from Embroidermodder to embroider's full test suite.)

The flag \texttt{--full-test-suite} runs all the tests that have been written. Since this results in a lot of output the details are both to stdout and to a text file called \texttt{test_matrix.txt}.

Patches that strictly improve the results in the \texttt{test_matrix.txt} over the current version will likely be accepted and it'll be a good place to go digging for contributions. (Note: strictly improve means that the testing result for each test is as good a result, if not better. Sacrificing one critera for another would require some design work before we would consider it.)

Symbols}

\index{symbols}

Symbols use the SVG path syntax.

In theory, we could combine the icons and symbols systems, since they could be rendered once and stored as icons in Qt. (Or as textures in FreeGLUT.)

Also we want to render the patterns themselves using SVG syntax, so it would save on repeated work overall.

Features}

Bindings}

\index{bindings}

Bindings for libembroidery are maintained for the languages we use internally in the project, for other languages we consider that the responsibility of other teams using the library.

So libembroidery is going to be supported on:

\begin{itemize} \item \texttt{C} (by default) \item \texttt{C++} (also by default) \item \texttt{Java} (for the Android\index{Android} application MobileViewer) \item \texttt{Swift} (for the iOS\index{iOS} application iMobileViewer) \end{itemize}

For \texttt{C\#}\index{C-sharp} we recommend directly calling the function directly using the DllImport feature:

\begin{lstlisting} /* Calling readCsv() via C# as a native function. */ [DllImport("libembroidery.so", EntryPoint="readCsv")] \end{lstlisting}

see this StackOverflow discussion for help: \url{https://stackoverflow.com/questions/11425202/is-it-possible-to-call-a-c-function-from-c-net}.

For Python you can do the same using ctypes: \url{https://www.geeksforgeeks.org/how-to-call-a-c-function-in-python/}.

Other Supported Thread Brands}

The thread lists that aren't preprogrammed into formats but are indexed in the data file for the purpose of conversion or fitting to images/graphics.

\begin{itemize} \item Arc Polyester \item Arc Rayon \item Coats and Clark Rayon \item Exquisite Polyester \item Fufu Polyester \item Fufu Rayon \item Hemingworth Polyester \item Isacord Polyester \item Isafil Rayon \item Marathon Polyester \item Marathon Rayon \item Madeira Polyester \item Madeira Rayon \item Metro Polyester \item Pantone \item Robison Anton Polyester \item Robison Anton Rayon \item Sigma Polyester \item Sulky Rayon \item ThreadArt Rayon \item ThreadArt Polyester \item ThreaDelight Polyester \item Z102 Isacord Polyester \end{itemize}

GUI Design}

Embroidermodder 2 was written in C++/Qt5 and it was far too complex. We had issues with people not able to build from source because the Qt5 libraries were so ungainly. So I decided to do a rewrite in C/SDL2 (originally FreeGLUT, but that was a mistake) with data stored as YAML. This means linking 4-7 libraries depending on your system which are all well supported and widely available.

This is going well, although it's slow progress as I'm trying to keep track of the design while also doing a ground up rewrite. I don't want to throw away good ideas. Since I also write code for libembroidery my time is divided. Overview of the UI rewrite

(Problems to be solved in brackets.)

It's not much to look at because I'm trying to avoid using an external widgets system, which in turn means writing things like toolbars and menubars over. If you want to get the design the actuator is the heart of it.

Without Qt5 we need a way of assigning signals with actions, so this is what I've got: the user interacts with a UI element, this sends an integer to the actuator that does the thing using the current state of the mainwindow struct of which we expect there to be exactly one instance. The action is taken out by a jump table that calls the right function (most of which are missing in action and not connected up properly). It also logs the number, along with key parts of the main struct in the undo history (an unsolved problem because we need to decide how much data to copy over per action). This means undo, redo and repeat actions can refer to this data.

To Do}

These should be sorted into relevant code sections.

\begin{itemize} \item todosort todo list. \item Alpha: High priority \begin{itemize} \item Statistics from 1.0, needs histogram \item Saving DST/PES/JEF (varga) \item Saving CSV/SVG (rt) + CSV read/write UNKNOWN interpreted as COLOR bug \end{itemize} \item Alpha: medium priority \begin{itemize} \item Notify user of data loss if not saving to an object format. \item Import Raster Image \item SNAP/ORTHO/POLAR \item Layer Manager + LayerSwitcher DockWidget \item Reading DXF \end{itemize} \item Alpha: low priority \begin{itemize} \item Writing DXF \item Up and Down keys cycle thru commands in the command prompt \item Amount of Thread, Machine Time Estimation (also allow customizable times for setup, color changes, manually trimming jump threads, etc…that way a realistic total time can be estimated) \item Otto Theme Icons - whatsthis icon doesn't scale well, needs redone \item embroidermodder2.ico 16 x 16 looks horrible \end{itemize} \item Alpha: lower priority \begin{itemize} \item CAD Command: Arc (rt) \end{itemize} \item beta \begin{itemize} \item Custom Filter Bug - doesn't save changes in some cases \item Cannot open file with \texttt{#} in name when opening multiple files (works fine when opening the single file) \item Closing Settings Dialog with the X in the window saves settings rather than discards them \item Advanced Printing \item Filling Algorithms (varga) \item Otto Theme Icons - beta (rt) - Units, Render, Selectors \end{itemize} \item Finish before 2.0 release \begin{itemize} \item QDoc Comments \item Review KDE4 Thumbnailer \item Documentation for libembroidery and formats \item HTML Help files \item Update language translations \item CAD Command review: line \item CAD Command review: circle \item CAD Command review: rectangle \item CAD Command review: polygon \item CAD Command review: polyline \item CAD Command review: point \item CAD Command review: ellipse \item CAD Command review: arc \item CAD Command review: distance \item CAD Command review: locatepoint \item CAD Command review: move \item CAD Command review: rgb \item CAD Command review: rotate \item CAD Command review: scale \item CAD Command review: singlelinetext \item CAD Command review: star \item Clean up all compiler warning messages, right now theres plenty :P \end{itemize} \item 2.0 \begin{itemize} \item tar.gz archive \item zip archive \item Debian Package (rt) \item NSIS Installer (rt) \item Mac Bundle? \item press release \end{itemize} \item 2.x/Ideas \begin{itemize} \item todo(2.x/ideas) libembroidery.mk for MXE project (refer to qt submodule packages for qmake based building. Also refer to plibc.mk for example of how write an update macro for github.) \item todo(2.x/ideas) libembroidery safeguard for all writers - check if the last stitch is an END stitch. If not, add an end stitch in the writer and modify the header data if necessary. \item todo(2.x/ideas) Cut/Copy - Allow Post-selection \item todo(2.x/ideas) CAD Command: Array \item todo(2.x/ideas) CAD Command: Offset \item todo(2.x/ideas) CAD Command: Extend \item todo(2.x/ideas) CAD Command: Trim \item todo(2.x/ideas) CAD Command: BreakAtPoint \item todo(2.x/ideas) CAD Command: Break2Points \item todo(2.x/ideas) CAD Command: Fillet \item todo(2.x/ideas) CAD Command: Chamfer \item todo(2.x/ideas) CAD Command: Split \item todo(2.x/ideas) CAD Command: Area \item todo(2.x/ideas) CAD Command: Time \item todo(2.x/ideas) CAD Command: PickAdd \item todo(2.x/ideas) CAD Command: Product \item todo(2.x/ideas) CAD Command: Program \item todo(2.x/ideas) CAD Command: ZoomFactor \item todo(2.x/ideas) CAD Command: GripHot \item todo(2.x/ideas) CAD Command: GripColor and GripCool \item todo(2.x/ideas) CAD Command: GripSize \item todo(2.x/ideas) CAD Command: Highlight \item todo(2.x/ideas) CAD Command: Units \item todo(2.x/ideas) CAD Command: Grid \item todo(2.x/ideas) CAD Command: Find \item todo(2.x/ideas) CAD Command: Divide \item todo(2.x/ideas) CAD Command: ZoomWindow (Move out of view.cpp) \item todo(2.x/ideas) Command: Web (Generates Spiderweb patterns) \item todo(2.x/ideas)

Command: Guilloche (Generates Guilloche patterns) \item todo(2.x/ideas) Command: Celtic Knots \item todo(2.x/ideas) Command: Knotted Wreath \item todo(2.x/ideas) Lego Mindstorms NXT/EV3 ports and/or commands. \item todo(2.x/ideas) native function that flashes the command prompt to get users attention when using the prompt is required for a command. \item todo(2.x/ideas) libembroidery-composer like app that combines multiple files into one. \item todo(2.x/ideas) Settings Dialog, it would be nice to have it notify you when switching tabs that a setting has been changed. Adding an Apply button is what would make sense for this to happen. \item todo(2.x/ideas) Keyboard Zooming/Panning \item todo(2.x/ideas) G-Code format? \item todo(2.x/ideas) 3D Raised Embroidery \item todo(2.x/ideas) Gradient Filling Algorithms \item todo(2.x/ideas) Stitching Simulation \item todo(2.x/ideas) RPM packages? \item todo(2.x/ideas) Reports? \item todo(2.x/ideas) Record and Playback Commands \item todo(2.x/ideas) Settings option for reversing zoom scrolling direction \item todo(2.x/ideas) Qt GUI for libembroidery-convert \item todo(2.x/ideas) EPS format? Look at using Ghostscript as an optional add-on to libembroidery… \item todo(2.x/ideas) optional compile option for including LGPL/GPL libs etc… with warning to user about license requirements. \item todo(2.x/ideas) Realistic Visualization - Bump Mapping/OpenGL/Gradients? \item todo(2.x/ideas) Stippling Fill \item todo(2.x/ideas) User Designed Custom Fill \item todo(2.x/ideas) Honeycomb Fill \item todo(2.x/ideas) Hilbert Curve Fill \item todo(2.x/ideas) Sierpinski Triangle fill \item todo(2.x/ideas) Circle Grid Fill \item todo(2.x/ideas) Spiral Fill \item todo(2.x/ideas) Offset Fill \item todo(2.x/ideas) Brick Fill \item todo(2.x/ideas) Trim jumps over a certain length. \item todo(2.x/ideas) FAQ about setting high number of jumps for more controlled trimming. \item todo(2.x/ideas) Minimum stitch length option. (Many machines also have this option too) \item todo(2.x/ideas) Add 'Design Details' functionality to libembroidery-convert \item todo(2.x/ideas) Add 'Batch convert many to one format' functionality to libembroidery-convert \item todo(2.x/ideas) EmbroideryFLOSS - Color picker that displays catalog numbers and names. \end{itemize} \item todo(beta) Realistic Visualization - Bump Mapping/OpenGL/Gradients? \item todo(beta) Get undo history widget back (BUG). \item todo(beta) Mac Bundle, .tar.gz and .zip source archive. \item todo(beta) NSIS installer for Windows, Debian package, RPM package \item todo(beta) GUI frontend for embroider features that aren't supported by embroidermodder: flag selector from a table \item todo(beta) Update all formats without color to check for edr or rgb files. \item todo(beta) Setting for reverse scrolling direction (for zoom, vertical pan) \item todo(beta) Keyboard zooming, panning \item todo(beta) New embroidermodder2.ico 16x16 logo that looks good at that scale. \item todo(beta) Saving dst, pes, jef. \item todo(beta) Settings dialog: notify when the user is switching tabs that the setting has been changed, adding apply button is what would make sense for this to happen. \item todo(beta) Update language translations. \item todo(beta) Replace KDE4 thumbnailer. \item todo(beta) Import raster image. \item todo(beta) Statistics from 1.0, needs histogram. \item todo(beta) SNAP/ORTHO/POLAR. \item todo(beta) Cut/copy allow post-selection. \item todo(beta) Layout into config. \item todo(beta) Notify user of data loss if not saving to an object format. \item todo(beta) Add which formats to work with to preferences. \item todo(beta) Cannot open file with \texttt{\#} in the name when opening multiple files but works with opening a single file. \item todo(beta) Closing settings dialog with the X in the window saves settings rather than discarding them. \item todo(beta) Otto theme icons: units, render, selectors, what's this icon doesn't scale. \item todo(beta) Layer manager and Layer switcher dock widget. \item todo(beta) Test that all formats read data in correct scale (format details should match other programs). \item todo(beta) Custom filter bug — doesn't save changes in some cases. \item todo(beta) Tools to find common problems in the source code and suggest fixes to the developers. For example, a translation miss: that is, for any language other than English a missing entry in the translation table should supply a clear warning to developers. \item todo(beta) Converting Qt C++ version to native GUI C throughout. \item todo(beta) OpenGL Rendering: \texttt{Real} rendering to see what the

embroidery looks like, Icons and toolbars, Menu bar. \item todo(beta) Libembroidery interfacing: get all classes to use the proper libembroidery types within them. So \texttt{Ellipse} has \texttt{EmbEllipse} as public data within it. \item todo(beta) Move calculations of rotation and scaling into \texttt{EmbVector} calls. \item todo(beta) GUI frontend for embroider features that aren't supported by embroidermodder: flag selector from a table \item todo(beta) Update all formats without color to check for edr or rgb files. \item todo(beta) Setting for reverse scrolling direction (for zoom, vertical pan) \item todo(beta) Keyboard zooming, panning \item todo(beta) Better integrated help: I don't think the help should backend to a html file somewhere on the user's system. A better system would be a custom widget within the program that's searchable. \item todo(beta) New embroidermodder2.ico 16x16 logo that looks good at that scale. \item todo(beta) Settings dialog: notify when the user is switching tabs that the setting has been changed, adding apply button is what would make sense for this to happen. \end{itemize}

Contributing}

Version Control}

Being an open source project, developers can grab the latest code at any time and attempt to build it themselves. We try our best to ensure that it will build smoothly at any time, although occasionally we do break the build. In these instances, please provide a patch, pull request which fixes the issue or open an issue and notify us of the problem, as we may not be aware of it and we can build fine.

Try to group commits based on what they are related to: features/bugs/comments/graphics/commands/etc...

See the coding style here (coding-style).

Get the Development Build going}

When we switch to releases we recommend using them, unless you're reporting a bug in which case you can check the development build for whether it has been patched. If this applies to you, the current development build is:

\begin{itemize} \item Linux (https://github.com/Embroidermodder/Embroidermodder/suites/8882922866/artifacts/406005099) \item Mac OS (https://github.com/Embroidermodder/Embroidermodder/suites/8882922866/artifacts/406005101) \item Windows (https://github.com/Embroidermodder/Embroidermodder/suites/8882922866/artifacts/406005102) \end{itemize}

\begin{itemize} \item todo(beta) Libembroidery 1.0.

\item todo(beta) Better integrated help: I don't think the help should backend to a html file somewhere on the user's system. A better system would be a custom widget within the program that's searchable.

\item todo(beta) EmbroideryFLOSS - Color picker that displays catalog numbers and names.

\item todo(beta) Custom filter bug — doesn't save changes in some cases.

\item todo(beta) Advanced printing.

\item todo(beta) Stitching simulation.

\item todo(2.x/ideas) User designed custom fill. \end{itemize}

These are key bits of reasoning behind why the GUI is built the way it is.

Translation of the user interface}

In a given table the left column is the default symbol and the right string is the translation. If the translate function fails to find a translation it returns the default symbol.

So in US English it is an empty table, but in UK English only the dialectical differences are present.

Ideally, we should support at least the 6 languages spoken at the UN. Quoting www.un.org:

\begin{quote} \emph{There are six official languages of the UN. These are Arabic, Chinese, English, French, Russian and Spanish.} \end{quote}

We're adding Hindi, on the grounds that it is one of the most commonly spoken languages and at least one of the Indian languages should be present.

Written Chinese is generally supported as two different symbol sets and we follow that convension.

English is supported as two dialects to ensure that the development team is aware of what those differences are. The code base is written by a mixture of US and UK native English speakers meaning that only the variable names are consistently one dialect: US English. As for documentation: it is whatever dialect the writer prefers (but they should maintain consistency within a text block like this one).

Finally, we have "default", which is the dominant language of the internals of the software. Practically, this is just US English, but in terms of programming history this is the "C locale".

Old action system notes}

Action: the basic system to encode all user input.

This typedef gives structure to the data associated with each action which, in the code, is referred to by the action id (an int from the define table above).

DESCRIPTION OF STRUCT CONTENTS}

label}

The action label is always in US English, lowercase, seperated with hyphens.

For example: \texttt{new-file}.

Flags}

The bit based flags all collected into a 32-bit integer.

\begin{longtable}{p{3cm} p{10cm}} \caption{Flags of EM actions} \label{tab:flags-for-actions}\\ bit(s) & description \\ \hline 0 & User (0) or system (1) permissions. \\ 1-3 & The mode of input. \\ 4-8 & The object classes that this action can be applied to. \\ 9-10 & What menu (if any) should it be present in. \\ 11-12 & What \end{longtable}

Description}

The string placed in the tooltip describing the action.

\section{Original Prompt System}

NOTE	\texttt{main()} is run every time the command is started. Use it to reset variables so they are ready to go.
NOTE	\texttt{click()} is run only for left clicks. Middle clicks are used for panning. Right clicks bring up the context menu.
NOTE	\texttt{move()} is optional. It is run only after \texttt{enableMoveRapidFire()} is called. It will be called every time the mouse moves until \texttt{disableMoveRapidFire()} is called.
NOTE	\texttt{prompt()} is run when Enter is pressed. \texttt{appendPromptHistory} is automatically called before \texttt{prompt()} is called so calling it is only needed for erroneous input. Any text in the command prompt is sent as an uppercase string.

CAD command review}

\index{newfile} \index{openfile} \index{savefile} \index{scale} \index{circle} \index{offset} \index{extend} \index{trim} \index{break-at-point} \index{break-2-points} \index{fillet} \index{star}

\begin{longtable}{p{1cm} p{2cm} p{3cm} p{5cm}} \textbf{ID} & \textbf{name} & \textbf{arguments} & \textbf{description} \\ \hline 0 & \texttt{newfile} & none & Create a new EmbPattern with a new tab in the GUI. \\ 1 & \texttt{openfile} & filename string & Open an EmbPattern with the supplied filename \texttt{fname}. \\ 2 & \texttt{savefile} & filename string & Save the current loaded EmbPattern to the supplied filname \texttt{fname}. \\ 3 & \texttt{scale} & selected objects, 1 float & Scale all selected objects by the number supplied, without selection scales the entire design \\ 4 & \texttt{circle} & mouse co-ords & Adds a circle to the design based on the supplied numbers, converts to stitches on save for stitch only formats. \\ 5 & \texttt{offset} & mouse co-ords & Shifts the selected objects by the amount given by the mouse co-ordinates. \\ 6 & \texttt{extend} & & \\ 7 & \texttt{trim} & & \\ 8 & \texttt{break_at_point} & & \\ 9 & \texttt{break_2_points} & & \\ 10 & \texttt{fillet} & & \\ 11 & \texttt{star} & & \\ 12 & \texttt{singlelinetext} \index{singlelinetext} & & \\ 13 & \index{chamfer} \texttt{chamfer} & & \\ 14 & \index{split} \texttt{split} & & \\ 15 & \index{area}

\texttt{area} & & \\ 16 & \index{time} \texttt{time} & & \\ 17 & \index{pickadd} \texttt{pickadd} & & \\ 16 & \index{zoomfactor} \texttt{zoomfactor} & & \\ 17 & \index{product} \texttt{product} & & \\ 18 & \index{program} \texttt{program} & & \\ 19 & \index{zoomwindow} \texttt{zoomwindow} & & \\ 20 & \index{divide} \texttt{divide} & & \\ 21 & \index{find} \texttt{find} & & \\ 22 & \index{record} \texttt{record} & & \\ 23 & \index{playback} \texttt{playback} & & \\ 24 & \index{rotate} \texttt{rotate} & & \\ 25 & \index{rgb} \texttt{rgb} & & \\ 26 & \index{move} \texttt{move} & & \\ 27 & \index{grid} \texttt{grid} & & \\ 28 & \index{griphot} \texttt{griphot} & & \\ 29 & \index{gripcolor} \texttt{gripcolor} & & \\ 30 & \index{gripcool} \texttt{gripcool} & & \\ 31 & \index{gripsize} \texttt{gripsize} & & \\ 32 & \index{highlight} \texttt{highlight} & & \\ 33 & \index{units} \texttt{units} & & \\ 34 & \index{locatepoint} \texttt{locatepoint} & & \\ 35 & \index{distance} \texttt{distance} & & \\ 36 & \index{arc} \texttt{arc} & & \\ 37 & \index{ellipse} \texttt{ellipse} & & \\ 38 & \index{array} \texttt{array} & & \\ 39 & \index{point} \texttt{point} & & \\ 40 & \index{polyline} \texttt{polyline} & & \\ 41 & \index{polygon} \texttt{polygon} & & \\ 42 & \index{rectangle} \texttt{rectangle} & & \\ 43 & \index{line} \texttt{line} & & \\ 44 & \index{arc-rt} \texttt{arc (rt)} & & \\ 45 & \index{dolphin} \texttt{dolphin} & & \\ 46 & \index{heart} \texttt{heart} & & \end{longtable}

Actions}

ARC}

\index{arc}

CIRCLE}

\index{circle}

OPEN}

\index{open}

\section{Changelog}

\section{Ideas}

Stuff that is now supposed to be generated by Doxygen:

\begin{itemize} \item todo: Bibliography style to plainnat. \item todo: US letter paper version of printed docs. \end{itemize}

\chapter{Formats}

\section{Overview}

Read/Write Support Levels}

The table of read/write format support levels uses the status levels described here:

\begin{longtable}{p{3cm} p{7cm}} Status Label & Description \\ \hline \texttt{rw-none} & Either the format produces no output, reporting an error. Or it produces a Tajima dst file as an alternative. \\

\texttt{rw-poor} & A file somewhat similar to our examples is produced. We don't know how well it runs on machines in practice as we don't have any user reports or personal tests. \\ \texttt{rw-basic} & Simple files in this format run well on machines that use this format. \\ \texttt{rw-standard} & Files with non-standard features work on machines and we have good documentation on the format. \\ \texttt{rw-reliable} & All known features don't cause crashes. Almost all work as expected. \\ \texttt{rw-complete} & All known features of the format work on machines that use this format. Translations from and to this format preserve all features present in both. \end{longtable}

These can be split into \texttt{r-basic w-none}, for example, if they don't match.

So all formats can, in principle, have good read and good write support, because it's defined in relation to files that we have described the formats for.

Test Support Levels}

\begin{longtable}{p{3cm} p{7cm}} Status Label & Description \\ \hline \texttt{test-none} & No tests have been written to test the specifics of the format. \\ \texttt{test-basic} & Stitch Lists and/or colors have read/write tests. \\ \texttt{test-thorough} & All features of that format has at least one test. \\ \texttt{test-fuzz} & Can test the format for uses of features that we haven't thought of by feeding in nonsense that is designed to push possibly dangerous weaknesses to reveal themselves. \\ \texttt{test-complete} & Both thorough and fuzz testing is covered. \end{longtable}

So all formats can, in principle, have complete testing support, because it's defined in relation to files that we have described the formats for.

Documentation Support Levels}

\begin{longtable}{p{3cm} p{7cm}} Status Label & Description \\ \hline \texttt{doc-none} & We haven't researched this beyond finding example files. \\ \texttt{doc-basic} & We have a rough sketch of the size and contents of the header if there is one. We know the basic stitch encoding (if there is one), but not necessarily all stitch features. \\ \texttt{doc-standard} & We know some good sources and/or have tested all the features that appear to exist. They mostly work the way we have described. \\ \texttt{doc-good} & All features that were described somewhere have been covered here or we have thoroughly tested our ideas against other softwares and hardwares and they work as expected. \\ \texttt{doc-complete} & There is a known official description and our description covers all the same features. \end{longtable}

Not all formats can have complete documentation because it's based on what information is publically available. So the total score is reported in the table below based on what level we think is available.

Overall Support}

Since the overall support level is the combination of these 4 factors, but rather than summing up their values it's an issue of the minimum support of the 4.

\begin{longtable}{p{3cm} p{7cm}} Status Label & Description \\ \hline \texttt{read-only} & If write support is none and read support is not none. \\ \texttt{write-only} & If read support is none and write support is not none. \\ \texttt{unstable} & If both read and write support are not none but

testing or documentation is none. \\ \texttt{basic} & If all ratings are better than none. \\ \texttt{reliable} & If all ratings are better than basic. \\ \texttt{complete} & If all ratings could not reasonably be better (for example any improvements rely on information that we may never have access to). This is the only status that can be revoked, since if the format changes or new documentation is released it is no longer complete. \\ \texttt{experimental} & For all other scenarios. \end{longtable}

Table of Format Support Levels}

Overview of documentation support by format.

\begin{longtable}{p{5cm} p{3cm} p{3cm}} Format & Ratings & Score \\ \hline Toyota Embroidery Format (.100) & rw-basic doc-none test-none & unstable \\ Toyota Embroidery Format (.10o) & rw-basic doc-none test-none & unstable \\ Bernina Embroidery Format (.art) & rw-none doc-none test-none & experimental \\ Bitmap Cache Embroidery Format (.bmc) & r-basic w-none doc-none test-none & unstable \\ Bits and Volts Embroidery Format (.bro) & rw-none doc-none test-none & experimental \\ Melco Embroidery Format (.cnd) & rw-none doc-none test-none & experimental \\ Embroidery Thread Color Format (.col) & rw-basic doc-none test-none & \texttt{experimental} \\ Singer Embroidery Format (.csd) & rw-none doc-none test-none & experimental \\ Comma Separated Values (.csv) & rw-none doc-none test-none & experimental \\ Barudan Embroidery Format (.dat) & rw-none doc-none test-none & experimental \\ Melco Embroidery Format (.dem) & rw-none doc-none test-none & experimental \\ Barudan Embroidery Format (.dsb) & rw-none doc-none test-none & experimental \\ Tajima Embroidery Format (.dst) & rw-none doc-none test-none & experimental \\ ZSK USA Embroidery Format (.dsz) & rw-none doc-none test-none & experimental \\ Drawing Exchange Format (.dxf) & rw-none doc-none test-none & experimental \\ Embird Embroidery Format (.edr) & rw-none doc-none test-none & experimental \\ Elna Embroidery Format (.emd) & rw-none doc-none test-none & experimental \\ Melco Embroidery Format (.exp) & rw-none doc-none test-none & experimental \\ Eltac Embroidery Format (.exy) & rw-none doc-none test-none & experimental \\ Sierra Expanded Embroidery Format (.eys) & rw-none doc-none test-none & experimental \\ Fortron Embroidery Format (.fxy) & rw-none doc-none test-none & experimental \\ Smoothie G-Code Embroidery Format (.gc) & rw-none doc-none test-none & experimental \\ Great Notions Embroidery Format (.gnc) & rw-none doc-none test-none & experimental \\ Gold Thread Embroidery Format (.gt) & rw-none doc-none test-none & experimental \\ Husqvarna Viking Embroidery Format (.hus) & rw-none doc-none test-none & experimental \\ Inbro Embroidery Format (.inb) & rw-none doc-none test-none & experimental \\ Embroidery Color Format (.inf) & rw-none doc-none test-none & experimental \\ Janome Embroidery Format (.jef) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.ksm) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.max) & rw-none doc-none test-none & experimental \\ Mitsubishi Embroidery Format (.mit) & rw-none doc-none test-none & experimental \\ Ameco Embroidery Format (.new) & rw-none doc-none test-none & experimental \\ Melco Embroidery Format (.ofm) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.pcd) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.pcm) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.pcq) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.pcs) & rw-none doc-none test-none & experimental \\ Brother Embroidery Format (.pec) & rw-none doc-none test-none & experimental \\ Brother Embroidery Format (.pel) & rw-none doc-none test-none & experimental \\ Brother Embroidery Format (.pem) & rw-none doc-none test-none & experimental \\ Brother Embroidery Format (.pes) & rw-none doc-none test-none & experimental \\ Brother Embroidery Format (.phb) & rw-none doc-

none test-none & experimental \\ Brother Embroidery Format (.phc) & rw-none doc-none test-none & experimental \\ AutoCAD Embroidery Format (.plt) & rw-none doc-none test-none & experimental \\ RGB Embroidery Format (.rgb) & rw-none doc-none test-none & experimental \\ Janome Embroidery Format (.sew) & rw-none doc-none test-none & experimental \\ Husqvarna Viking Embroidery Format (.shv) & rw-none doc-none test-none & experimental \\ Sunstar Embroidery Format (.sst) & rw-none doc-none test-none & experimental \\ Data Stitch Embroidery Format (.stx) & rw-none doc-none test-none & experimental \\ Scalable Vector Graphics (.svg) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.t01) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.t09) & rw-none doc-none test-none & experimental \\ Happy Embroidery Format (.tap) & rw-none doc-none test-none & experimental \\ ThredWorks Embroidery Format (.thr) & rw-none doc-none test-none & experimental \\ Text File (.txt) & rw-none doc-none test-none & experimental \\ Barudan Embroidery Format (.u00) & rw-none doc-none test-none & experimental \\ Barudan Embroidery Format (.u01) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.vip) & rw-none doc-none test-none & experimental \\ Pfaff Embroidery Format (.vp3) & rw-none doc-none test-none & experimental \\ Singer Embroidery Format (.xxx) & rw-none doc-none test-none & experimental \\ ZSK USA Embroidery Format (.zsk) & rw-none doc-none test-none & experimental \end{longtable}

Format Support}

\index{10o} \index{100} \index{art} \index{bro} \index{cnd} \index{col} \index{csd} \index{dat} \index{dem} \index{dsb} \index{dst} \index{dsz} \index{dxf} \index{edr} \index{emd} \index{exp} \index{exy} \index{fxy} \index{gnc} \index> \index{hus} \index{inb} \index{jef} \index{ksm} \index{pcd} \index{pcm} \index{pcq} \index{pcs} \index{pec} \index{pel} \index{pem} \index{pes} \index{phb} \index{phc} \index{rgb} \index{sew} \index{shv} \index{sst} \index{svg} \index{tap} \index{u01} \index{vip} \index{vp3} \index{xxx} \index{zsk}

\begin{longtable}{p{2cm} p{2cm} p{2cm} p{5cm}} FORMAT & READ & WRITE & NOTES \\ \hline \texttt{10o} & YES & & read (need to fix external color loading) (maybe find out what ctrl and code flags of 0x10, 0x08, 0x04, and 0x02 mean) \\ \texttt{100} & & & none (4 byte codes) 61 00 10 09 (type, type2, x, y ?) x and y (signed char) \\ \texttt{100} & & & none (4 byte codes) 61 00 10 09 (type, type2, x, y ?) x and y (signed char) \\ \texttt{art} & & & none \\ \texttt{bro} & YES & & read (complete)(maybe figure out detail of header) \\ \texttt{cnd} & & & none \\ \texttt{col} & & & (color file no design) read(final) write(final) \\ \texttt{csd} & YES & & read (complete) \\ \texttt{dat} & & & read () \\ \texttt{dem} & & & none (looks like just encrypted cnd) \\ \texttt{dsb} & YES & & read (unknown how well) (stitch data looks same as 10o) \\ \texttt{dst} & YES & & read (complete) / write(unknown) \\ \texttt{dsz} & YES & & read (unknown) \\ \texttt{dxf} & & & read (Port to C. needs refactored) \\ \texttt{edr} & & & read (C version is broken) / write (complete) \\ \texttt{emd} & & & read (unknown) \\ \texttt{exp} & YES & & read (unknown) / write(unknown) \\ \texttt{exy} & YES & & read (need to fix external color loading) \\ \texttt{fxy} & YES & & read (need to fix external color loading) \\ \texttt{gnc} & & & none \\ \texttt> & & & read (need to fix external color loading) \\ \texttt{hus} & YES & & read (unknown) / write (C version is broken) \\ \texttt{inb} & YES & & read (buggy?) \\ \texttt{jef} & YES & & write (need to fix the offsets when it is moving to another spot) \\ \texttt{ksm} & YES & & read (unknown) / write (unknown) \\ \texttt{pcd} & & & \\ \texttt{pcm} & & & \\ \texttt{pcq} & & & read (Port to C) \\ \texttt{pcs} & BUGGY & & read (buggy / colors are not correct / after reading, writing any other format is messed up) \\ \texttt{pec} & & & read / write (without embedded images, sometimes overlooks some stitches leaving a gap) \\ \texttt{pel} & & & none \\ \texttt{pem} & & & none \\ \texttt{pes} & YES & & \\ \texttt{phb} & & & \\ \texttt{phc} & & & \\

\texttt{rgb} & & & \\ \texttt{sew} & YES & & \\ \texttt{shv} & & & read (C version is broken) \\ \texttt{sst} & & & none \\ \texttt{svg} & & YES & \\ \texttt{tap} & YES & & read (unknown) \\ \texttt{u01} & & & \\ \texttt{vip} & YES & & \\ \texttt{vp3} & YES & & \\ \texttt{xxx} & YES & & \\ \texttt{zsk} & & & read (complete) \end{longtable}

\begin{itemize} \item TODO Josh, Review this section and move any info still valid or needing work into TODO comments in the actual libembroidery code. Many items in this list are out of date and do not reflect the current status of libembroidery. When finished, delete this file. \begin{itemize} \item Test that all formats read data in correct scale (format details should match other programs) \item Add which formats to work with to preferences. \item Check for memory leaks \item Update all formats without color to check for edr or rgb files \item Fix issues with DST (VERY important that DST work well) \end{itemize} \item todoSupport for Singer FHE, CHE (Compucon) formats? \end{itemize}

\chapter{Geometry and Algorithms}

\section{To Do}

Arduino}

\begin{itemize} \item Fix emb-outline files \item Fix thread-color files \item Logging of Last Stitch Location to External USB Storage(commonly available and easily replaced) ...wait until TRE is available to avoid rework \item inotool.org - seems like the logical solution for Nightly/CI builds \item Smoothieboard experiments \end{itemize}

Testing}

\begin{itemize} \item looping test that reads 10 times while running valgrind. See \texttt{embPattern_loadExternalColorFile()} Arduino leak note for more info. \end{itemize}

Development}

If you wish to develop with us you can chat via the contact email on the website \url{https://libembroidery.org} or in the issues tab on the github page \url{https://github.com/Embroidermodder/Embroidermodder/issues}. People have been polite and friendly in these conversations and I (Robin) have really enjoyed them. If we do have any arguments please note we have a Code of Conduct CODE_OF_CONDUCT.md so there is a consistent policy to enforce when dealing with these arguments.

The first thing you should try is building from source using the build advice (build) above. Then read some of the manual \url{https://libembroidery.org/embroidermodder_2.0_manual.pdf} to get the general layout of the source code and what we are currently planning.

Testing}

To find unfixed errors run the tests by launching from the command line with:

\begin{lstlisting} $ embroidermodder --test \end{lstlisting}

then dig through the output. It's currently not worth reporting the errors, since there are so many but if you can fix anything reported here you can submit a PR.

Contributing}

Funding}

The easiest way to help is to fund development (see the Donate button above), since we can't afford to spend a lot of time developing and only have limited kit to test out libembroidery on.

Programming and Engineering}

Should you want to get into the code itself:

\begin{itemize} \item Low level C developers are be needed for the base library \texttt{libembroidery}. \item Low level assembly programmers are needed for translating some of \texttt{libembroidery} to \texttt{EmbroiderBot}. \item Hardware Engineers to help design our own kitbashed embroidery machine \texttt{EmbroiderBot}, one of the original project aims in 2013. \item Scheme developers and C/SDL developers to help build the GUI. \item Scheme developers to help add designs for generating of custom stitch-filled emblems like the heart or dolphi. Note that this happens in Embroidermodder not libembroidery (which assumes that you already have a function available). \end{itemize}

Writing}

We also need people familiar with the software and the general machine embroidery ecosystem to contribute to the documentation (https://github.com/Embroidermodder/docs).

We need researchers to find references for the documentation: colour tables, machine specifications etc. The history is murky and often very poorly maintained so if you know anything from working in the industry that you can share: it'd be appreciated!

Embroidermodder Project Coding Standards}

A basic set of guidelines to use when submitting code.

Code structure is mre important than style, so first we advise you read ``Design'' and experimenting before getting into the specifics of code style.

Where Code Goes}

Anything that deals with the specifics of embroidery file formats, threads, rendering to images, embroidery machinery or command line interfaces should go in \texttt{libembroidery} not here.

Where Non-compiled Files Go}

\todo{Like most user interfaces Embroidermodder is mostly data, so here we will have a list describing where each CSV goes.}

Ways in which we break style on purpose}

Most style guides advise you to keep functions short. We make a few pointed exceptions to this where the overall health and functionality of the source code should benefit.

The \texttt{actuator} function will always be a mess and it should be: we're keeping the total source lines of code down by encoding all user action into a descrete sequence of strings that are all below \texttt{_STRING_LENGTH} in length. See the section on the actuator (TODO) describing why any other solution we could think here would mean more more code without a payoff in speed of execution or clarity.

Version Control}

Being an open source project, developers can grab the latest code at any time and attempt to build it themselves. We try our best to ensure that it will build smoothly at any time, although occasionally we do break the build. In these instances, please provide a patch, pull request which fixes the issue or open an issue and notify us of the problem, as we may not be aware of it and we can build fine.

Try to group commits based on what they are related to: features/bugs/comments/graphics/commands/etc…

Donations}

Creating software that interfaces with hardware is costly. A summary of some of the costs involved:

\begin{itemize} \item Developer time for 2 core developers \item Computer equipment and parts \item Embroidery machinery \item Various electronics for kitbashing Embroiderbot \item Consumable materials (thread, fabric, stabilizer, etc…) \end{itemize}

If you have found our software useful, please consider funding further development by donating to the project on Open Collective (\url{https://opencollective.com/embroidermodder}).

Embroidermodder Project Coding Standards}

Rather than maintain our own standard for style, please defer to the Python's PEP 7 \citet{pep7} for C style and emulating that in C++.

A basic set of guidelines to use when submitting code. Defer to the PEP7 standard with the following additions:

\begin{itemize} \item All files and directories shall be lowercase and contain no spaces. \item Structs and class names should use \texttt{LeadingCapitals}. \item Enums and constants should be \texttt{BLOCK_CAPITALS}. \item Class members and functions without a parent class should be \texttt{snake_case}. With the exception of when one of the words is a "class" name from libembroidery in which case it has the middle capitals like this: \texttt{embArray_add}. \item Don't use exceptions. \item Don't use ternary operator (?:) in place of if/else. \item Don't repeat a variable name that already occurs in an outer scope. \end{itemize}

Version Control}

Being an open source project, developers can grab the latest code at any time and attempt to build it themselves. We try our best to ensure that it will build smoothly at any time, although occasionally we do break the build. In these instances, please provide a patch, pull request which fixes the issue or open an issue and notify us of the problem, as we may not be aware of it and we can build fine.

Try to group commits based on what they are related to: features/bugs/comments/graphics/commands/etc…

Comments}

When writing code, sometimes there are items that we know can be improved, incomplete or need special clarification. In these cases, use the types of comments shown below. They are pretty standard and are highlighted by many editors to make reviewing code easier. We also use shell scripts to parse the code to find all of these occurrences so someone wanting to go on a bug hunt will be able to easily see which areas of the code need more love.

libembroidery and Embroidermodder are written in C and adheres to C89 standards. This means that any C99 or C++ comments will show up as errors when compiling with gcc. In any C code, you must use:

\begin{lstlisting} /* Use C Style Comments within code blocks. * * Use Doxygen style code blocks to place todo, bug, hack, warning, * and note items like this: * * \todo EXAMPLE: This code clearly needs more work or further review. * * \bug This code is definitely wrong. It needs fixed. * * \hack This code shouldn't be written this way or I don't * feel right about it. There may a better solution * * \warning Think twice (or more times) before changing this code. * I put this here for a good reason. * * \note This comment is much more important than lesser comments. */ \end{lstlisting}

Ideas}

Why this document}

I've been trying to make this document indirectly through the Github issues page and the website we're building but I think a straightforward, plain-text file needs to be the ultimate backup for this. Then I can have a printout while I'm working on the project.

Qt and dependencies}

I'm switching to SDL2 (which is a whole other conversation) which means we can ship it with the source code package meaning only a basic build environment is necessary to build it.

Documentation}

Can we treat the website being a duplicate of the docs a non-starter? I'd be happier with tex/pdf only and (I know this is counter-intuitive) one per project.

Social Platform}

So… all the issues and project boards etc. being on Github is all well and good assuming that we have our own copies. But we don't if Github goes down or some other major player takes over the space and we have to move (again, since this started on SourceForge).

This file is a backup for that which is why I'm repeating myself between them.

Identify the meaning of these TODO items}

\begin{itemize} \item Saving CSV/SVG (rt) + CSV read/write UNKNOWN interpreted as COLOR bug \texttt{\#179} \item Lego Mindstorms NXT/EV3 ports and/or commands \end{itemize}

Progress Chart}

The chart of successful from-to conversions (previously a separate issue) is something that should appear in the README.

Standard}

The criteria for a good Pull Request from an outside developer has these properties, from most to least important:

\begin{itemize} \item No regressions on testing. \item Add a feature, bug fix or documentation that is already agreed on through GitHub issues or some other way with a core developer. \item No GUI specific code should be in libembroidery, that's for Embroidermodder. \item Pedantic/ansi C unless there's a good reason to use another language. \item Meet the style above (i.e. PEP 7, Code Lay-out (\url{https://peps.python.org/pep-0007/#code-lay-out})). We'll just fix the style if the code's good and it's not a lot of work. \item \texttt{embroider} should be in POSIX style as a command line program. \item No dependancies that aren't "standard", i.e. use only the C Standard Library. \end{itemize}

Image Fitting}

A currently unsolved problem in development that warrants further research is the scenario where a user wants to feed embroider an image that can then be .

To Place}

A \emph{right-handed coordinate system}\index{right-handed coordinate system} is one where up is positive and right is positive. Left-handed is up is positive, left is positive. Screens often use down is positive, right is positive, including the OpenGL standard so when switching between graphics formats and stitch formats we need to use a vertical flip (\texttt{embPattern_flip}).

\texttt{0x20} is the space symbol, so when padding either 0 or space is preferred and in the case of space use the literal ' '.

To Do}

We currently need help with:

\begin{itemize} \item Thorough descriptions of each embroidery format. \item Finding resources for each of the branded thread libraries (along with a full citation for documentation). \item Finding resources for each geometric algorithm used (along with a full citation for documentation). \item Completing the full \texttt{--full-test-suite} with no segfaults and at least a clear error message (for example `not implemented yet''). \item Identifying best guesses'' for filling in missing information when going from, say \texttt{.csv} to a late \texttt{.pes} version. What should the default be when the data doesn't clarify? \item Improving the written documentation. \item Funding, see the Sponsor button above. We can treat this as ``work'' and put far more hours in with broad support in small donations from people who want specific features. \end{itemize}

Beyond this the development targets are categories sorted into:

\begin{itemize} \item Basic Features \item Code quality and user friendliness \item embroider CLI \item Documentation \item GUI \item electronics development \end{itemize}

Basic features}

\begin{itemize} \item Incorporate \texttt{\#if 0}ed parts of \texttt{libembroidery.c}. \item Interpret how to write formats that have a read mode from the source code and vice versa. \item Document the specifics of the file formats here for embroidery machine specific formats. Find websites and other sources that break down the binary formats we currently don't understand. \item Find more and better documentation of the structure of the headers for the formats we do understand. \end{itemize}

Code quality and user friendliness}

\begin{itemize} \item Document all structs, macros and functions (will contribute directly on the web version). \item Incorporate experimental code, improve support for language bindings. \item Make stitch x, y into an EmbVector. \end{itemize}

Documentation}

Run \texttt{sloccount} on \texttt{extern/} and \texttt{.} (and) so we know the current scale of the project, aim to get this number low. Report the total as part of the documentation.

Try to get as much of the source code that we maintain into C as possible so new developers don't need to learn multiple languages to have an effect. This bars the embedded parts of the code.

GUI}

\begin{itemize} \item Make EmbroideryMobile (Android) also backend to \texttt{libembroidery} with a Java wrapper. \item Make EmbroideryMobile (iOS) also backend to \texttt{libembroidery} with a Swift wrapper. \item Share some of the MobileViewer and iMobileViewer layout with the main EM2. Perhaps combine those 3 into the Embroidermodder repository so there are 4 repositories total. \item Convert layout data to JSON format and use cJSON for parsing. \end{itemize}

Development}

Contributing}

If you're interested in getting involved, here's some guidance for new developers. Currently The Embroidermodder Team is all hobbyists with an interest in making embroidery machines more open and user friendly. If you'd like to support us in some other way you can donate to our Open Collective page (click the Donate button) so we can spend more time working on the project.

All code written for libembroidery should be ANSI C89 compliant if it is C. Using other languages should only be used where necessary to support bindings.

Debug}

If you wish to help with development, run this debug script and send us the error log.

\begin{lstlisting} #!/bin/bash

rm -fr libembroidery-debug

git clone http://github.com/embroidermodder/libembroidery libembroidery-debug cd libembroidery-debug

cmake -DCMAKE_BUILD_TYPE=DEBUG . cmake --build . --config=DEBUG

valgrind ./embroider --full-test-suite \end{lstlisting}

While we will attempt to maintain good results from this script as part of normal development it should be the first point of failure on any system we haven't tested or format we understand less.

Binary download

We need a current \texttt{embroider} command line program download, so people can update without building.

Programming principles for the C core

End arrays of char arrays with the symbol "END", the code will never require this symbol as an entry.

Define an array as one of 3 kinds: constant, editable or data.

- Constant arrays are defined const and have fixed length matching the data.
- Editable arrays are fixed length, but to a length based on the practical use of that array. A dropdown menu can't contain more than 30 items, because we don't want to flood the user with options. However it can nest indefinately, so it is not restricted to a total number of entries.
- Data arrays is editable and changes total size at runtime to account for user data.

Style rules for arrays

1.

Libembroidery on Embedded Systems

The libembroidery library is designed to support embedded environments as well as desktop, so it can be used in CNC applications.

Originally, the embedded system aspect of the Embroidermodder project was targeted at the higher end Arduino\index{Arduino} prototyping board as part of a general effort to make our own open source hardware (OSHW)\index{OSHM}.

However, the task of building the interface for a full OSHW embroidery machine neatly splits into the tasks of building a user interface to issue the commands and the rig itself starting with the stepper motors that wire into this control circuit. A well built control circuit could issue commands to a variety of different machine layouts (for example many features are not present on some machines)

Compatible Boards

We recommend using an Arduino\index{Arduino} Mega 2560 or another board with equal or greater specs. That being said, we have had success using an Arduino Uno R3 but this will likely require further optimization and other improvements to ensure continued compatibility with the Uno. See below for more information.

Arduino Considerations

There are two main concerns here: Flash Storage and SRAM.

libembroidery continually outgrows the 32KB of Flash storage on the Arduino Uno and every time this occurs, a decision has to be made as to what capabilities should be included or omitted. While reading files is the main focus on arduino, writing files may also play a bigger role in the future. Long term, it would be most practical to handle the inclusion or omission of any feature via a single configuration header file that the user can modify to suit their needs.

SRAM is in extremely limited supply and it will deplete quickly so any dynamic allocation should occur early during the setup phase of the sketch and sparingly or not at all later in the sketch. To help minimize SRAM consumption on Arduino and ensure libembroidery can be used in any way the sketch creator desires, it is required that any sketch using libembroidery must implement event handlers. See the ino-event source and header files for more information.

There is also an excellent article by Bill Earl on the Adafruit Learning System[^1].

[^1]: http://learn.adafruit.com/memories-of-an-arduino?view=all

Space

Since a stitch takes 3 bytes of storage and many patterns use more than 10k stitches, we can't assume that the pattern will fit in memory. Therefore we will need to buffer the current pattern on and off storage in small chunks. By the same reasoning, we can't load all of one struct beore looping so we will need functions similar to binaryReadInt16 for each struct.

This means the EmbArray approach won't work since we need to load each element and dynamic memory management is unnecessary because the arrays lie in storage.

TODO: Replace EmbArray functions with embPattern load functions.

Tables

All thread tables and large text blocks are too big to compile directly into the source code. Instead we can package the library with a data packet that is compiled from an assembly program in raw format so the specific padding can be controlled.

In the user section above we will make it clear that this file needs to be loaded on the pattern USB/SD card or the program won't function.

TODO: Start file with a list of offsets to data with a corresponding table to load into with macro constants for each label needed.

Current Pattern Memory Management

It will be simpler to make one file per EmbArray so we keep an EmbFile* and a length, so no malloc call is necessary. So there needs to be a consistent tmpfile naming scheme.

TODO: For each pattern generate a random string of hexadecimal and append it to the filenames like \texttt{stitchList_A16F.dat}. Need to check for a file which indicates that this string has been used already.

Special Notes

Due to historical reasons and to remain compatible with the Arduino 1.0 IDE, this folder must be called ``utility''. Refer to the arduino build process for more info[^2].

[^2]: https://arduino.github.io/arduino-cli/0.19/sketch-build-process/

libembroidery relies on the Arduino SD library for reading files. See the ino-file source and header files for more information.

The Assembly Split

One problem to the problem of supporting both systems with abundant memory (such as a 2010s or

later desktop) and with scarce memory (such as embedded systems) is that they don't share the same assembly language. To deal with this: there will be two equivalent software which are hand engineered to be similar but one will be in C and the other in the assembly dialects we support.

All assembly will be intended for embedded systems only, since a slightly smaller set of features will be supported. However, we will write a \texttt{x86} version since that can be tested.

That way the work that has been done to simplify the C code can be applied to the assembly versions.

Electronics development

Ideas

Currently experimenting with Fritzing 8 (8), upload netlists to embroiderbot when they can run simulations using the asm in `libembroidery`.

Create a common assembly for data that is the same across chipsets `libembrodiery\_data\_internal.s`.

Make the defines part of `embroidery.h` all systems and the function list `c code only`. That way we can share some development between assembly and C versions.

\include{cli.tex}

Mobile

Again, it would help to use the C library we have already developed, however for Android the supported platform is for Java applications.

https://github.com/java-native-access/jna

https://github.com/marketplace/actions/setup-android-ndk

https://developer.android.com/ndk/guides

See the bindings section for how this is achieved.

Tread Tables

Arc Polyester Codes

Red	Green	Blue	Name	Catalog Number
-	-	-	-	-

Arc Rayon Codes

Red	Green	Blue	Name	Catalog Number
-	-	-	-	-

Coats and Clark Rayon Codes

Red	Green	Blue	Name	Catalog Number
-	-	-	-	-

title = "Coats and Clark Code" author = "The Embroidermodder Team" abbrev = "coats-and-clark" date = "2023-10-02"

Coats and Clark Rayon Codes

| Red | Green | Blue | Name | Catalog Number | |-----|-------|------|------|----------------| | - | - | - | - | - |

title = "DXF Codes" author = "The Embroidermodder Team" abbrev = "dxf" date = "2023-10-02"

DXF Colors

Based on the DraftSight color table.

| Red | Green | Blue | Index | Name | |-----|-------|------|-------|------| | 0 | 0 | 0 | 0 | BYBLOCK | | 255 | 0 | 0 | 1 | red | | 255 | 255 | 0 | 2 | yellow | | 0 | 255 | 0 | 3 | green | | 0 | 255 | 255 | 4 | cyan | | 0 | 0 | 255 | 5 | blue | | 255 | 0 | 255 | 6 | magenta | | 255 | 255 | 255 | 7 | white | | 128 | 128 | 128 | 8 | dark gray | | 192 | 192 | 192 | 9 | light gray | | 255 | 0 | 0 | 10 | unknown | | 255 | 127 | 127 | 11 | unknown | | 204 | 0 | 0 | 12 | unknown | | | 204 | 102 | 102 | 13 | unknown | | 153 | 0 | 0 | 14 | unknown | | 153 | 76 | 76 | 15 | unknown | | 127 | 0 | 0 | 16 | unknown | | 127 | 63 | 63 | 17 | unknown | | 76 | 0 | 0 | 18 | unknown | | 76 | 38 | 38 | 19 | unknown | | 255 | 63 | 0 | 20 | unknown | | 255 | 159 | 127 | 21 | unknown | | 204 | 51 | 0 | 22 | unknown | | 204 | 127 | 102 | 23 | unknown | | 153 | 38 | 0 | 24 | unknown | | 153 | 95 | 76 | 25 | unknown | | 127 | 31 | 0 | 26 | unknown | | 127 | 79 | 63 | 27 | unknown | | 76 | 19 | 0 | 28 | unknown | | 76 | 47 | 38 | 29 | unknown | | 255 | 127 | 0 | 30 | unknown | | 255 | 191 | 127 | 31 | unknown | | 204 | 102 | 0 | 32 | unknown | | 204 | 153 | 102 | 33 | unknown | | 153 | 76 | 0 | 34 | unknown | | 153 | 114 | 76 | 35 | unknown | | 127 | 63 | 0 | 36 | unknown | | 127 | 95 | 63 | 37 | unknown | | 76 | 38 | 0 | 38 | unknown | | 76 | 57 | 38 | 39 | unknown | | 255 | 191 | 0 | 40 | unknown | | 255 | 223 | 127 | 41 | unknown | | 204 | 153 | 0 | 42 | unknown | | 204 | 178 | 102 | 43 | unknown | | 153 | 114 | 0 | 44 | unknown | | 153 | 133 | 76 | 45 | unknown | | 127 | 95 | 0 | 46 | unknown | | 127 | 111 | 63 | 47 | unknown | | 76 | 57 | 0 | 48 | unknown | | 76 | 66 | 38 | 49 | unknown | | 255 | 255 | 0 | 50 | unknown | | 255 | 255 | 127 | 51 | unknown | | 204 | 204 | 0 | 52 | unknown | | 204 | 204 | 102 | 53 | unknown | | 153 | 153 | 0 | 54 | unknown | | 153 | 153 | 76 | 55 | unknown | | 127 | 127 | 0 | 56 | unknown | | 127 | 127 | 63 | 57 | unknown | | 76 | 76 | 0 | 58 | unknown | | 76 | 76 | 38 | 59 | unknown | | 191 | 255 | 0 | 60 | unknown | | 223 | 255 | 127 | 61 | unknown

| | 153 | 204 | 0 | 62 | unknown | | 178 | 204 | 102 | 63 | unknown | | 114 | 153 | 0 | 64 | unknown | | 133 | 153 | 76 | 65 | unknown | | 95 | 127 | 0 | 66 | unknown | | 111 | 127 | 63 | 67 | unknown | | 57 | 76 | 0 | 68 | unknown | | 66 | 76 | 38 | 69 | unknown | | 127 | 255 | 0 | 70 | unknown | | 191 | 255 | 127 | 71 | unknown | | 102 | 204 | 0 | 72 | unknown | | 153 | 204 | 102 | 73 | unknown | | 76 | 153 | 0 | 74 | unknown | | 114 | 153 | 76 | 75 | unknown | | 63 | 127 | 0 | 76 | unknown | | 95 | 127 | 63 | 77 | unknown | | 38 | 76 | 0 | 78 | unknown | | 57 | 76 | 38 | 79 | unknown | | 63 | 255 | 0 | 80 | unknown | | 159 | 255 | 127 | 81 | unknown | | 51 | 204 | 0 | 82 | unknown | | 127 | 204 | 102 | 83 | unknown | | 38 | 153 | 0 | 84 | unknown | | 95 | 153 | 76 | 85 | unknown | | 31 | 127 | 0 | 86 | unknown | | 79 | 127 | 63 | 87 | unknown | | 19 | 76 | 0 | 88 | unknown | | 47 | 76 | 38 | 89 | unknown | | 0 | 255 | 0 | 90 | unknown | | 127 | 255 | 127 | 91 | unknown | | 0 | 204 | 0 | 92 | unknown | | 102 | 204 | 102 | 93 | unknown | | 0 | 153 | 0 | 94 | unknown | | 76 | 153 | 76 | 95 | unknown | | 0 | 127 | 0 | 96 | unknown | | 63 | 127 | 63 | 97 | unknown | | 0 | 76 | 0 | 98 | unknown | | 38 | 76 | 38 | 99 | unknown | | 0 | 255 | 63 | 100 | unknown | | 127 | 255 | 159 | 101 | unknown | | 0 | 204 | 51 | 102 | unknown | | 102 | 204 | 127 | 103 | unknown | | 0 | 153 | 38 | 104 | unknown | | 76 | 153 | 95 | 105 | unknown | | 0 | 127 | 31 | 106 | unknown | | 63 | 127 | 79 | 107 | unknown | | 0 | 76 | 19 | 108 | unknown | | 38 | 76 | 47 | 109 | unknown | | 0 | 255 | 127 | 110 | unknown | | 127 | 255 | 191 | 111 | unknown | | 0 | 204 | 102 | 112 | unknown | | 102 | 204 | 153 | 113 | unknown | | 0 | 153 | 76 | 114 | unknown | | 76 | 153 | 114 | 115 | unknown | | 0 | 127 | 63 | 116 | unknown | | 63 | 127 | 95 | 117 | unknown | | 0 | 76 | 38 | 118 | unknown | | 38 | 76 | 57 | 119 | unknown | | 0 | 255 | 191 | 120 | unknown | | 127 | 255 | 223 | 121 | unknown | | 0 | 204 | 153 | 122 | unknown | | 102 | 204 | 178 | 123 | unknown | | 0 | 153 | 114 | 124 | unknown | | 76 | 153 | 133 | 125 | unknown | | 0 | 127 | 95 | 126 | unknown | | 63 | 127 | 111 | 127 | unknown | | 0 | 76 | 57 | 128 | unknown | | 38 | 76 | 66 | 129 | unknown | | 0 | 255 | 255 | 130 | unknown | | 127 | 255 | 255 | 131 | unknown | | 0 | 204 | 204 | 132 | unknown | | 102 | 204 | 204 | 133 | unknown | | 0 | 153 | 153 | 134 | unknown | | 76 | 153 | 153 | 135 | unknown | | 0 | 127 | 127 | 136 | unknown | | 63 | 127 | 127 | 137 | unknown | | 0 | 76 | 76 | 138 | unknown | | 38 | 76 | 76 | 139 | unknown | | 0 | 191 | 255 | 140 | unknown | | 127 | 223 | 255 | 141 | unknown | | 0 | 153 | 204 | 142 | unknown | | 102 | 178 | 204 | 143 | unknown | | 0 | 114 | 153 | 144 | unknown | | 76 | 133 | 153 | 145 | unknown | | 0 | 95 | 127 | 146 | unknown | | 63 | 111 | 127 | 147 | unknown | | 0 | 57 | 76 | 148 | unknown | | 38 | 66 | 76 | 149 | unknown | | 0 | 127 | 255 | 150 | unknown | | 127 | 191 | 255 | 151 | unknown | | 0 | 102 | 204 | 152 | unknown | | 102 | 153 | 204 | 153 | unknown | | 0 | 76 | 153 | 154 | unknown | | 76 | 114 | 153 | 155 | unknown | | 0 | 63 | 127 | 156 | unknown | | 63 | 95 | 127 | 157 | unknown | | 0 | 38 | 76 | 158 | unknown | | 38 | 57 | 76 | 159 | unknown | | 0 | 63 | 255 | 160 | unknown | | 127 | 159 | 255 | 161 | unknown | | 0 | 51 | 204 | 162 | unknown | | 102 | 127 | 204 | 163 | unknown | | 0 | 38 | 153 | 164 | unknown | | 76 | 95 | 153 | 165 | unknown | | 0 | 31 | 127 | 166 | unknown | | 63 | 79 | 127 | 167 | unknown | | 0 | 19 | 76 | 168 | unknown | | 38 | 47 | 76 | 169 | unknown | | 0 | 0 | 255 | 170 | unknown | | 127 | 127 | 255 | 171 | unknown | | 0 | 0 | 204 | 172 | unknown | | 102 | 102 | 204 | 173 | unknown | | 0 | 0 | 153 | 174 | unknown | | 76 | 76 | 153 | 175 | unknown | | 0 | 0 | 127 | 176 | unknown | | 63 | 63 | 127 | 177 | unknown | | 0 | 0 | 76 | 178 | unknown | | 38 | 38 | 76 | 179 | unknown | | 63 | 0 | 255 | 180 | unknown | | 159 | 127 | 255 | 181 | unknown | | 51 | 0 | 204 | 182 | unknown | | 127 | 102 | 204 | 183 | unknown | | 38 | 0 | 153 | 184 | unknown | | 95 | 76 | 153 | 185 | unknown | | 31 | 0 | 127 | 186 | unknown | | 79 | 63 | 127 | 187 | unknown | | 19 | 0 | 76 | 188 | unknown | | 47 | 38 | 76 | 189 | unknown | | 127 | 0 | 255 | 190 | unknown | | 191 | 127 | 255 | 191 | unknown | | 102 | 0 | 204 | 192 | unknown | | 153 | 102 | 204 | 193 | unknown | | 76 | 0 | 153 | 194 | unknown | | 114 | 76 | 153 | 195 | unknown | | 63 | 0 | 127 | 196 | unknown | | 95 | 63 | 127 | 197 | unknown | | 38 | 0 | 76 | 198 | unknown | | 57 | 38 | 76 | 199 | unknown | | 191 | 0 |

255 | 200 | unknown | | 223 | 127 | 255 | 201 | unknown | | 153 | 0 | 204 | 202 | unknown | | 178 | 102 | 204 | 203 | unknown | | 114 | 0 | 153 | 204 | unknown | | 133 | 76 | 153 | 205 | unknown | | 95 | 0 | 127 | 206 | unknown | | 111 | 63 | 127 | 207 | unknown | | 57 | 0 | 76 | 208 | unknown | | 66 | 38 | 76 | 209 | unknown | | 255 | 0 | 255 | 210 | unknown | | 255 | 127 | 255 | 211 | unknown | | 204 | 0 | 204 | 212 | unknown | | 204 | 102 | 204 | 213 | unknown | | 153 | 0 | 153 | 214 | unknown | | 153 | 76 | 153 | 215 | unknown | | 127 | 0 | 127 | 216 | unknown | | 127 | 63 | 127 | 217 | unknown | | 76 | 0 | 76 | 218 | unknown | | 76 | 38 | 76 | 219 | unknown | | 255 | 0 | 191 | 220 | unknown | | 255 | 127 | 223 | 221 | unknown | | 204 | 0 | 153 | 222 | unknown | | 204 | 102 | 178 | 223 | unknown | | 153 | 0 | 114 | 224 | unknown | | 153 | 76 | 133 | 225 | unknown | | 127 | 0 | 95 | 226 | unknown | | 127 | 63 | 111 | 227 | unknown | | 76 | 0 | 57 | 228 | unknown | | 76 | 38 | 66 | 229 | unknown | | 255 | 0 | 127 | 230 | unknown | | 255 | 127 | 191 | 231 | unknown | | 204 | 0 | 102 | 232 | unknown | | 204 | 102 | 153 | 233 | unknown | | 153 | 0 | 76 | 234 | unknown | | 153 | 76 | 114 | 235 | unknown | | 127 | 0 | 63 | 236 | unknown | | 127 | 63 | 95 | 237 | unknown | | 76 | 0 | 38 | 238 | unknown | | 76 | 38 | 57 | 239 | unknown | | 255 | 0 | 63 | 240 | unknown | | 255 | 127 | 159 | 241 | unknown | | 204 | 0 | 51 | 242 | unknown | | 204 | 102 | 127 | 243 | unknown | | 153 | 0 | 38 | 244 | unknown | | 153 | 76 | 95 | 245 | unknown | | 127 | 0 | 31 | 246 | unknown | | 127 | 63 | 79 | 247 | unknown | | 76 | 0 | 19 | 248 | unknown | | 76 | 38 | 47 | 249 | unknown | | 51 | 51 | 51 | 250 | unknown | | 91 | 91 | 91 | 251 | unknown | | 132 | 132 | 132 | 252 | unknown | | 173 | 173 | 173 | 253 | unknown | | 214 | 214 | 214 | 254 | unknown | | 255 | 255 | 255 | 255 | unknown | | 0 | 0 | 0 | 256 | BYLAYER |

thread_color Exquisite_Polyester_codes[] = { {"END", 0, -1} };

Fufu Polyester Codes

| Red | Green | Blue | Name | Catalog Code | |-----|-------|------|------|----------| | - | - | - | - | - |

Fufu Rayon Codes

| Red | Green | Blue | Name | Catalog Code | |----|--------|-------|-------|----------| | - | - | - | - | - |

thread_color Hemingworth_Polyester_codes[] = { {"Pure White", 0xFFFFFFFF, 1001}, {"Lemon Ice", 0xFFDDE00F, 1271}, {"Neon Green", 0xFFC9DD03, 1272}, {"Brilliant Lime", 0xFF60DD49, 1273}, {"Mango", 0xFFFFCC1E, 1274}, {"Neon Yellow", 0xFFFFED38, 1275}, {"Tropical Orange", 0xFFFFA952, 1276}, {"Neon Orange", 0xFFFF9338, 1277}, {"Rebel Peach", 0xFFFF585F, 1278}, {"Shy Flamingo", 0xFFF28CA3, 1279}, {"Neon Pink", 0xFFFE8A9E, 1280}, {"Neon Peach", 0xFFFC074F, 1281}, {"Dove Gray", 0xFFCFC3C3, 1067}, {"Silver Lining", 0xFFC9CAC8, 1068}, {"Storm Cloud", 0xFFB2B4B3, 1069}, {"Platinum", 0xFFC6C6BC, 1070}, {"Graphite", 0xFF616365, 1244}, {"Light Charcoal", 0xFF4D4F53, 1245}, {"Chrome", 0xFF8E908F, 1072}, {"Antique Silver", 0xFF747678, 1077}, {"Pewter Gray", 0xFF6C6F70, 1073}, {"Black Stallion", 0xFF191D1F, 1079}, {"Charcoal", 0xFF1B242A, 1087}, {"Classic Black", 0xFF000000, 1000}, {"Marshmallow", 0xFFD5D2CA, 1118}, {"Ice Blue", 0xFFDAE3EA, 1119}, {"Nautical Blue", 0xFFA6BCC6, 1076}, {"Sea Storm", 0xFF818A8F, 1074}, {"Bronzed Steel", 0xFF595A5C, 1078}, {"Silvery Gray", 0xFFB9C9D0, 1239}, {"Granite", 0xFF7D9AAA, 1240}, {"Shadow", 0xFF5E6A71, 1085}, {"Dark Slate Blue", 0xFF003C69, 1241}, {"Deep Slate Blue", 0xFF003946, 1242}, {"Pacific Waters", 0xFF004250, 1081}, {"Dark Slate", 0xFF37424A, 1086}, {"Smoky Blue", 0xFF005B82, 1192}, {"Light Slate Blue", 0xFF5E9CAE, 1193}, {"Hyacinth", 0xFF6AADE4, 1188}, {"Bluebird",

0xFF4B92DB, 1187}, {"Misty Blue", 0xFF8FCAE7, 1186}, {"Cornflower Blue", 0xFFC2DEEA, 1185}, {"Pale Blue", 0xFFA0CFEB, 1256}, {"Country Blue", 0xFF0098DB, 1255}, {"Azure", 0xFF3D7EDB, 1202}, {"Royal Blue", 0xFF0039A6, 1203}, {"Brilliant Blue", 0xFF00338D, 1204}, {"Deep Blue", 0xFF0B2265, 1205}, {"Winter Blue", 0xFF98C6EA, 1189}, {"Winter Sky", 0xFFAACAE6, 1190}, {"Sky Blue", 0xFF8EBAE5, 1191}, {"China Blue", 0xFF0073CF, 1198}, {"Dark Blueberry", 0xFF004165, 1201}, {"Salem Blue", 0xFF004153, 1200}, {"Navy", 0xFF002244, 1199}, {"Sailor Blue", 0xFF002C5F, 1265}, {"Dark Blue ", 0xFF002857, 1264}, {"Berry Blue", 0xFF003591, 1263}, {"True Blue", 0xFF002C77, 1261}, {"Periwinkle", 0xFF6F9AD3, 1262}, {"Iceberg Blue", 0xFF65CFE9, 1197}, {"Medium Aquamarine", 0xFF0075B0, 1195}, {"Dark Aquamarine", 0xFF0066A1, 1196}, {"Peacock Blue", 0xFF006983, 1194}, {"Dark Turquoise", 0xFF003D4C, 1258}, {"Turquoise", 0xFF0098C3, 1259}, {"Caribbean Blue", 0xFF00B0CA, 1260}, {"Summer Sky", 0xFF6FD4E4, 1257}, {"Crystal Lake", 0xFFBBE7E6, 1178}, {"Icicle Blue", 0xFFC1E2E5, 1172}, {"Frosty Blue", 0xFF8FDFE2, 1173}, {"Blue Lagoon", 0xFF00AFD8, 1174}, {"Blue Satin", 0xFF006778, 1181}, {"Teal Blue", 0xFF007C92, 1180}, {"Light Teal Blue", 0xFF009AA6, 1176}, {"Wintergreen", 0xFF7CA295, 1175}, {"Mint Green", 0xFF63CECA, 1177}, {"Navajo Turquoise", 0xFF00877C, 1179}, {"Peacock Green", 0xFF007B69, 1182}, {"Forest Glen", 0xFF024E43, 1183}, {"Deep Teal", 0xFF004953, 1184}, {"Deep Sea", 0xFF156570, 1082}, {"Dragonfly", 0xFF00505C, 1113}, {"Blue Steel", 0xFF44697D, 1084}, {"Dark Sage", 0xFF496C60, 1114}, {"Silver Green", 0xFF949D9E, 1115}, {"Antique Gray", 0xFF91BAA3, 1243}, {"Ocean Spray", 0xFFB9CCC3, 1100}, {"Sea Foam", 0xFFA6E6BC, 1088}, {"Cucumber Melon", 0xFF00B588, 1094}, {"Light Jade", 0xFF00985F, 1106}, {"Jade", 0xFF009B74, 1107}, {"Dark Jade", 0xFF007D57, 1105}, {"Caribbean", 0xFF006A4D, 1104}, {"Dark Teal", 0xFF00685B, 1254}, {"Minty Teal", 0xFF0D776E, 1253}, {"Lemony Lime", 0xFFC3E76F, 1099}, {"Kiwi Lime", 0xFFCCDC00, 1247}, {"Electric Green", 0xFF69BE28, 1097}, {"Green Apple", 0xFF92D400, 1091}, {"Key Lime", 0xFF7AB800, 1092}, {"Kelly Green", 0xFF3F9C35, 1093}, {"Meadow", 0xFF00AF3F, 1248}, {"Grassy Green", 0xFF007934, 1095}, {"Dark Kelly Green", 0xFF008542, 1108}, {"Christmas Green", 0xFF00693C, 1109}, {"Winter Pine ", 0xFF1C453B, 1250}, {"Holly Leaf", 0xFF175E54, 1249}, {"Pistachio Nut", 0xFFC8E59A, 1096}, {"Dusty Green", 0xFF69923A, 1251}, {"Bush Ivy ", 0xFF557630, 1252}, {"Leafy Green", 0xFF739600, 1089}, {"Kentucky Grass", 0xFF53682B, 1090}, {"Ivy", 0xFF035642, 1103}, {"Evergreen", 0xFF284E36, 1110}, {"Mountain Meadow", 0xFF004438, 1111}, {"Forest Green", 0xFF004D46, 1112}, {"Oregano", 0xFF57584F, 1121}, {"Jungle Green", 0xFF404A29, 1123}, {"Thyme", 0xFF83847A, 1120}, {"Light Avocado", 0xFF827C34, 1058}, {"Split Pea", 0xFFB19B00, 1061}, {"Spring Leaf", 0xFFB5A300, 1059}, {"Almond Cream", 0xFFF8E498, 1060}, {"Eggshell", 0xFFD7D3C7, 1229}, {"Cornsilk Green", 0xFFD5C833, 1098}, {"Avocado", 0xFF6A7029, 1101}, {"Seaweed", 0xFF898F4B, 1102}, {"Olive Green", 0xFF65551C, 1246}, {"Coconut Shell", 0xFF4B452C, 1117}, {"Parsley", 0xFF4B471A, 1116}, {"Dried Sage", 0xFF718674, 1083}, {"Mocha", 0xFF4F4C25, 1125}, {"Warm Earth", 0xFF5D4F4B, 1131}, {"Dark Chocolate", 0xFF452325, 1126}, {"Deep Walnut", 0xFF4E2E2D, 1128}, {"Teddybear Brown", 0xFF6E3219, 1130}, {"Light Chestnut", 0xFF60351D, 1134}, {"Pecan Pie", 0xFF6C4D23, 1140}, {"Dark Alder", 0xFF766A65, 1237}, {"Army Green", 0xFF5B491F, 1137}, {"Pharaoh Gold", 0xFF6E5A2A, 1135}, {"Autumn Haystack", 0xFFAB8422, 1136}, {"Sahara", 0xFF856822, 1122}, {"Weathered Wood", 0xFF675C53, 1236}, {"Soft Beige", 0xFF9A996E, 1124}, {"Mercury", 0xFFC2B2B5, 1149}, {"Old Lace", 0xFFA5ACAF, 1146}, {"Caramel Cappuccino", 0xFFAE7D5B, 1145}, {"Brown Sugar", 0xFFA76F3E, 1133}, {"Light Cinnamon", 0xFFA25022, 1064}, {"Cinnamon", 0xFF86431E, 1238}, {"Apple Cider", 0xFFB2541A, 1163}, {"Indian Paintbrush", 0xFF9A3B26, 1144}, {"Rust", 0xFF833820, 1164}, {"Toasted Almond", 0xFF825C26, 1142}, {"Pale Caramel", 0xFFC59217, 1063}, {"Honey Butter", 0xFFDDB99A, 1062}, {"Sandy Shore", 0xFFD2C295, 1057}, {"Ecru", 0xFFC2C2A0, 1056}, {"Malt", 0xFFB3B38C, 1055}, {"Antique Lace", 0xFFC7B37F, 1054}, {"Champagne", 0xFFBD9271, 1171}, {"Butter Taffy", 0xFFB3995D, 1138}, {"Cream Soda", 0xFFCEA98C, 1235}, {"Conch Shell", 0xFFE39B6C, 1139}, {"New Penny", 0xFFBA6F2E, 1132}, {"Pumpkin Spice", 0xFFBB650E, 1141}, {"Soft Sunlight", 0xFFEBE8B1,

1042}, {"Lemon Drop", 0xFFEEEC83, 1043}, {"Daffodil", 0xFFF3EC7A, 1045}, {"Lemon Citrus", 0xFFF5EC5A, 1225}, {"Sunshine Yellow", 0xFFFAE700, 1226}, {"Canary Yellow", 0xFFF2EE72, 1044}, {"Sunflower", 0xFFFCD900, 1227}, {"Sun", 0xFFFADC41, 1046}, {"Dandelion", 0xFFFED100, 1047}, {"Buttercup", 0xFFF3CF45, 1048}, {"Ginger Root", 0xFFEFBD47, 1050}, {"Goldenrod", 0xFFEAAB00, 1051}, {"Cornsilk", 0xFFDCD6B2, 1037}, {"Macadamia", 0xFFC6BC89, 1038}, {"Yellow Plumeria", 0xFFF8E498, 1039}, /* TODO: duplicate case value */ {"Maize", 0xFFF8DE6E, 1040}, {"Dried Banana", 0xFFFADA63, 1049}, {"Butternut", 0xFFFFCB4F, 1053}, {"Orange Meringue", 0xFFFFA100, 1232}, {"September Sunset", 0xFFFFB612, 1231}, {"Orange Cream", 0xFFFFB652, 1230}, {"Cantaloupe", 0xFFFFBC3D, 1041}, {"Old Gold", 0xFFCE8E00, 1052}, {"Auburn", 0xFF9D5324, 1143}, {"Citrus Burst", 0xFFE98300, 1024}, {"Orange Slice", 0xFFFF7000, 1025}, {"Fiery Sunset", 0xFFE37222, 1027}, {"Hunter Orange", 0xFFFB4F14, 1028}, {"Fall Harvest", 0xFFDD4814, 1029}, {"Candy Apple", 0xFFCD202C, 1030}, {"Christmas Red", 0xFFC30014, 1270}, {"Pomegranate", 0xFFA70232, 1032}, {"Rummy Raisin", 0xFF882332, 1031}, {"Cardinal Red", 0xFFA51100, 1002}, {"Rusty Red", 0xFF9E3039, 1234}, {"Redwood", 0xFF783014, 1233}, {"Carrot", 0xFFD55C19, 1065}, {"Paprika", 0xFFAA272F, 1066}, {"Cherrywood", 0xFF5F3327, 1129}, {"Burnt Sienna", 0xFF5D3526, 1127}, {"Merlot", 0xFF592C35, 1160}, {"Loganberry", 0xFF6A1A41, 1159}, {"Cranberry", 0xFF6E2714, 1158}, {"Mulberry", 0xFF662046, 1157}, {"Magenta", 0xFF85003C, 1156}, {"Raspberry", 0xFF641F14, 1155}, {"Salmon", 0xFFFF818D, 1166}, {"Georgia Peach", 0xFFFFA48A, 1015}, {"Rose Sunset", 0xFFFFB0B7, 1011}, {"Bubblegum Pink", 0xFFF3789B, 1012}, {"Carnation", 0xFFDB4D69, 1014}, {"Very Berry", 0xFF91004B, 1013}, {"Raspberry Red", 0xFF82240C, 1224}, {"Dark Salmon", 0xFFF54359, 1018}, {"Apricot Dream", 0xFFF4587A, 1017}, {"Coral Reef", 0xFFFF8B7C, 1016}, {"Frosted Peach", 0xFFFFC19C, 1022}, {"Tangerine", 0xFFFF8F70, 1020}, {"Dark Mango", 0xFFFF6D42, 1026}, {"Marigold", 0xFFFFA02F, 1023}, {"Spun Silk", 0xFFECC182, 1168}, {"Whipped Papaya", 0xFFFBCE92, 1021}, {"Baby Peach", 0xFFFDC480, 1228}, {"Pink Pearl", 0xFFFFC2A2, 1167}, {"Peaches 'n Cream", 0xFFEFC5CE, 1169}, {"Peach Pastel", 0xFFEFBE9C, 1170}, {"Old Penny", 0xFF774A39, 1162}, {"Dusty Rose", 0xFFB26F7E, 1151}, {"Winter Rose", 0xFFD490A8, 1165}, {"Valentine Pink", 0xFFF6A3BB, 1161}, {"Petal Peach", 0xFFFFB7AE, 1019}, {"Soft Petal", 0xFFEFD6DB, 1150}, {"Fuchsia", 0xFF920075, 1036}, {"Pink Kiss", 0xFFF375C6, 1004}, {"Baby Pink", 0xFFF3BBCE, 1003}, {"Whisper Pink", 0xFFF1DBDF, 1005}, {"Gentle Blush", 0xFFF3C9D3, 1006}, {"English Rose", 0xFFF4B2C1, 1007}, {"Sweet Pea", 0xFFF39EBC, 1008}, {"Rosy Blush", 0xFFF77AB4, 1009}, {"Passion Pink", 0xFFD71F85, 1010}, {"Mulled Wine", 0xFF772059, 1035}, {"Primrose", 0xFFC50084, 1034}, {"Azalea", 0xFFA1006B, 1033}, {"Snowflake", 0xFFD1D4D3, 1148}, {"Moonlight", 0xFFCAD1E7, 1147}, {"Tulip", 0xFF9DABE2, 1207}, {"Purple Iris", 0xFF8884D5, 1206}, {"Grape", 0xFF1A2155, 1209}, {"Moon Shadow", 0xFF6459C4, 1211}, {"Electric Purple", 0xFF212492, 1208}, {"Indigo", 0xFF411244, 1210}, {"Royal Purple", 0xFF3B0083, 1223}, {"Eggplant", 0xFF151C54, 1267}, {"Dark Purple", 0xFF490E6F, 1269}, {"Pure Purple", 0xFF57068C, 1268}, {"Pale Orchid", 0xFFDCC7DF, 1217}, {"Dusty Mauve", 0xFFC2ACBE, 1219}, {"Orchid", 0xFFDC9DDD, 1218}, {"Heather", 0xFFB382C7, 1213}, {"Lavender", 0xFF9C5FB5, 1214}, {"Soft Grape", 0xFF4B08A1, 1266}, {"Freesia", 0xFFC1AFE5, 1221}, {"Lilac", 0xFFC5B9E3, 1222}, {"Peony", 0xFF6E2C6B, 1215}, {"Dark Fuschia", 0xFF7D0063, 1216}, {"Grape Jelly", 0xFF752864, 1220}, {"Deep Orchid", 0xFF55517B, 1080}, {"Misty Blue Gray", 0xFF5C7F92, 1075}, {"Iron Ore", 0xFFAFADC3, 1071}, {"Light Mauve", 0xFFD8AAB3, 1152}, {"Dark Mauve", 0xFF89687C, 1153}, {"Wild Plum", 0xFF644459, 1154}, {"Huckleberry", 0xFF4B306A, 1212}, {"END", 0, -1} };

HUS Colors (find a citation)

; CSV format: red, green, blue, name, catalog_number ; 0, 0, 0 "Black",

"TODO:HUS_CATALOG_NUMBER" 0, 0, 255 "Blue", "TODO:HUS_CATALOG_NUMBER" 0, 255, 0 "Light Green", "TODO:HUS_CATALOG_NUMBER" 255, 0, 0 "Red", "TODO:HUS_CATALOG_NUMBER" 255, 0, 255 "Purple", "TODO:HUS_CATALOG_NUMBER" 255, 255, 0 "Yellow", "TODO:HUS_CATALOG_NUMBER" 127, 127, 127 "Gray", "TODO:HUS_CATALOG_NUMBER" 51, 154, 255 "Light Blue", "TODO:HUS_CATALOG_NUMBER" 51, 204, 102 "Green", "TODO:HUS_CATALOG_NUMBER" 255, 127, 0 "Orange", "TODO:HUS_CATALOG_NUMBER" 255, 160, 180 "Pink", "TODO:HUS_CATALOG_NUMBER" 153, 75, 0 "Brown", "TODO:HUS_CATALOG_NUMBER" 255, 255, 255 "White", "TODO:HUS_CATALOG_NUMBER" 0, 0, 127 "Dark Blue", "TODO:HUS_CATALOG_NUMBER" 0, 127, 0 "Dark Green", "TODO:HUS_CATALOG_NUMBER" 127, 0, 0 "Dark Red", "TODO:HUS_CATALOG_NUMBER" 255, 127, 127 "Light Red", "TODO:HUS_CATALOG_NUMBER" 127, 0, 127 "Dark Purple", "TODO:HUS_CATALOG_NUMBER" 255, 127, 255 "Light Purple", "TODO:HUS_CATALOG_NUMBER" 200, 200, 0 "Dark Yellow", "TODO:HUS_CATALOG_NUMBER" 255, 255, 153 "Light Yellow", "TODO:HUS_CATALOG_NUMBER" 60, 60, 60 "Dark Gray", "TODO:HUS_CATALOG_NUMBER" 192, 192, 192 "Light Gray", "TODO:HUS_CATALOG_NUMBER" 232, 63, 0 "Dark Orange", "TODO:HUS_CATALOG_NUMBER" 255, 165, 65 "Light Orange", "TODO:HUS_CATALOG_NUMBER" 255, 102, 122 "Dark Pink", "TODO:HUS_CATALOG_NUMBER" 255, 204, 204 "Light Pink", "TODO:HUS_CATALOG_NUMBER" 115, 40, 0 "Dark Brown", "TODO:HUS_CATALOG_NUMBER" 175, 90, 10 "Light Brown", "TODO:HUS_CATALOG_NUMBER" EOF

Isacord Polyester Codes

thread_color Isacord_Polyester_codes[] = { {"?", 0xFFFFFFFF, 10}, {"?", 0xFFFFFFFF, 15}, /* TODO: duplicate case value / **{"?", 0xFFFFFFFF, 17},** / TODO: duplicate case value / **{"?", 0xFF000000, 20}, {"?", 0xFFFFFDED, 101}, {"?", 0xFF6D757B, 108}, {"?", 0xFF515B61, 111}, {"?", 0xFF5D5D5D, 112}, {"?", 0xFFCFCFCF, 124}, {"?", 0xFFA1A9B4, 131}, {"?", 0xFF192024, 132}, {"?", 0xFF9EA5AA, 142}, {"?", 0xFFCFD1D5, 145}, {"?", 0xFFC6BDB4, 150}, {"?", 0xFFD5C4B3, 151}, {"?", 0xFF7C8283, 152}, {"?", 0xFFFEF5F0, 180}, {"?", 0xFFE9D7D9, 182}, {"?", 0xFFEBE3DD, 184}, {"?", 0xFFE0DA5F, 221}, {"?", 0xFFBFBA28, 232}, {"?", 0xFFFAF6CC, 250}, {"?", 0xFFF9F8E8, 270}, {"?", 0xFFFDF76C, 310}, {"?", 0xFFF5D300, 311}, {"?", 0xFF797E24, 345}, {"?", 0xFFB0AA76, 352}, {"?", 0xFF898F2B, 442}, {"?", 0xFF98996D, 453}, {"?", 0xFF6B7E6F, 463}, {"?", 0xFF3E4F34, 465}, {"?", 0xFFEDEF05, 501}, {"?", 0xFFF5D300, 506},** / TODO: duplicate case value / **{"?", 0xFFFDE896, 520}, {"?", 0xFFD7CE8A, 532}, {"?", 0xFFB18B00, 542}, {"?", 0xFFB28F11, 546}, {"?", 0xFFB69F56, 552}, {"?", 0xFFF8D73E, 600}, {"?", 0xFFF8D73E, 605},** / TODO: duplicate case value / **{"?", 0xFFF7DC00, 608}, {"?", 0xFFFEF09A, 630}, {"?", 0xFFFDE896, 640},** / TODO: duplicate case value / **{"?", 0xFFF5D2A6, 651}, {"?", 0xFFFEF9EA, 660}, {"?", 0xFFFAF6E7, 670}, {"?", 0xFFBEBEA8, 672}, {"?", 0xFFF7C35F, 700}, {"?", 0xFFF5CA00, 702}, {"?", 0xFFDFA200, 704}, {"?", 0xFFFCF538, 706}, {"?", 0xFFFADC59, 713}, {"?", 0xFF8C7E6A, 722}, {"?", 0xFF594900, 747}, {"?", 0xFFD6BF94, 761}, {"?", 0xFF656452, 776}, {"?", 0xFFF1AF00, 800}, {"?", 0xFFF5BA5D, 811}, {"?", 0xFFE1A23E, 821}, {"?", 0xFFCCAB3F, 822}, {"?", 0xFFDFA200, 824},** / TODO: duplicate case value / **{"?", 0xFFD0A44F, 832}, {"?", 0xFFCD944A, 842}, {"?", 0xFFE3BC61, 851}, {"?", 0xFF947C4A, 853}, {"?", 0xFFCBBFA2, 861}, {"?", 0xFFA5866A, 862}, {"?", 0xFFEBE7DD, 870}, {"?", 0xFF9FA086, 873}, {"?", 0xFF9A897B, 874}, {"?", 0xFFF3B259, 904}, {"?", 0xFFCA832C, 922}, {"?", 0xFFC07314, 931}, {"?", 0xFFAC6613, 932}, {"?", 0xFF744808, 933}, {"?", 0xFFBD9565, 934}, {"?", 0xFFC98300, 940}, {"?", 0xFFAF7D3E, 941}, {"?", 0xFF483928, 945}, {"?", 0xFFFEFEED, 970}, {"?", 0xFF6A4129, 1055}, {"?", 0xFFFDE2C1, 1060}, {"?", 0xFFA68A68, 1061}, {"?", 0xFFED9206, 1102}, {"?", 0xFFEE9C00, 1106}, {"?", 0xFFEE8751, 1114}, {"?", 0xFFA35214, 1115}, {"?",**

0xFFF8C000, 1120}, {"?", 0xFFB7976B, 1123}, {"?", 0xFF9D5A21, 1134}, {"?", 0xFFF3D8A8, 1140}, {"?", 0xFFFACFAE, 1141}, {"?", 0xFF7A4427, 1154}, {"?", 0xFFDFC8AB, 1172}, {"?", 0xFFE89763, 1211}, {"?", 0xFFF1A236, 1220}, {"?", 0xFFE5571D, 1300}, {"?", 0xFFD9674C, 1301}, {"?", 0xFFE4501E, 1304}, {"?", 0xFFEA7134, 1305}, {"?", 0xFFE12A23, 1306}, {"?", 0xFFC14817, 1311}, {"?", 0xFFC45331, 1312}, {"?", 0xFFD5815E, 1332}, {"?", 0xFFBB3D2E, 1334}, {"?", 0xFFBE2D1A, 1335}, {"?", 0xFF5F1B23, 1342}, {"?", 0xFF7A3441, 1346}, {"?", 0xFFFBBF95, 1351}, {"?", 0xFFF4A773, 1352}, {"?", 0xFF693920, 1355}, {"?", 0xFFF9C598, 1362}, {"?", 0xFF432731, 1366}, {"?", 0xFF464537, 1375}, {"?", 0xFFF4A782, 1430}, {"?", 0xFFE22D2A, 1501}, {"?", 0xFFA93121, 1514}, {"?", 0xFFEC7168, 1521}, {"?", 0xFFF6B08E, 1532}, {"?", 0xFFF9C5B9, 1551}, {"?", 0xFF806A61, 1565}, {"?", 0xFFE36C63, 1600}, {"?", 0xFFE44733, 1701}, {"?", 0xFFDF0032, 1703}, {"?", 0xFFE0003D, 1704}, {"?", 0xFFCF0040, 1725}, {"?", 0xFFF1CDCE, 1755}, {"?", 0xFFE9C9BD, 1760}, {"?", 0xFFE8C0B8, 1761}, {"?", 0xFFE00046, 1800}, {"?", 0xFFD6445D, 1805}, {"?", 0xFFF49E95, 1840}, {"?", 0xFFFCDAD5, 1860}, {"?", 0xFF636254, 1874}, {"?", 0xFF394535, 1876}, {"?", 0xFFE10057, 1900}, {"?", 0xFFBD0041, 1902}, {"?", 0xFFC00343, 1903}, {"?", 0xFFA9023A, 1904}, {"?", 0xFFBE004F, 1906}, {"?", 0xFF910230, 1911}, {"?", 0xFF86023E, 1912}, {"?", 0xFF9A0C3B, 1913}, {"?", 0xFFA33050, 1921}, {"?", 0xFFF28DA6, 1940}, {"?", 0xFFCE427A, 1950}, {"?", 0xFF959595, 1972}, {"?", 0xFFA33145, 2011}, {"?", 0xFF9F454C, 2022}, {"?", 0xFFC7979B, 2051}, {"?", 0xFF9F003F, 2101}, {"?", 0xFF78093F, 2113}, {"?", 0xFF6D0627, 2115}, {"?", 0xFF432732, 2123}, {"?", 0xFFE6778B, 2152}, {"?", 0xFFDF8390, 2153}, {"?", 0xFFF9BFC0, 2155}, {"?", 0xFFFBD1D6, 2160}, {"?", 0xFFD8D5D0, 2166}, {"?", 0xFFF7DED6, 2170}, {"?", 0xFFF7DEDE, 2171}, {"?", 0xFFE8418C, 2220}, {"?", 0xFF8C0C4A, 2222}, {"?", 0xFF883A40, 2224}, {"?", 0xFFEE71A1, 2230}, {"?", 0xFFA95A68, 2241}, {"?", 0xFFFAC8D3, 2250}, {"?", 0xFFD3007E, 2300}, {"?", 0xFFD20067, 2320}, {"?", 0xFF651533, 2333}, {"?", 0xFF3A212B, 2336}, {"?", 0xFFFDE5EC, 2363}, {"?", 0xFF970059, 2500}, {"?", 0xFFAA4381, 2504}, {"?", 0xFF820052, 2506}, {"?", 0xFFE20078, 2520}, {"?", 0xFFBF006A, 2521}, {"?", 0xFFF189AF, 2550}, {"?", 0xFFF7B4CA, 2560}, {"?", 0xFF494949, 2576}, {"?", 0xFF893480, 2600}, {"?", 0xFF6C0051, 2611}, {"?", 0xFFD994B9, 2640}, {"?", 0xFFE6B7CF, 2650}, {"?", 0xFFECD2DE, 2655}, {"?", 0xFF606D8C, 2674}, {"?", 0xFF610051, 2711}, {"?", 0xFF490251, 2715}, {"?", 0xFF89347F, 2720}, {"?", 0xFFC690A1, 2764}, {"?", 0xFF6F067B, 2810}, {"?", 0xFFA974AB, 2830}, {"?", 0xFF4C0F7B, 2900}, {"?", 0xFF664090, 2905}, {"?", 0xFF83589D, 2910}, {"?", 0xFF8C6DAA, 2920}, {"?", 0xFFC9B5D4, 3040}, {"?", 0xFFC790BA, 3045}, {"?", 0xFF707070, 3062}, {"?", 0xFF2A377E, 3102}, {"?", 0xFF35247A, 3110}, {"?", 0xFF260751, 3114}, {"?", 0xFF353A90, 3210}, {"?", 0xFF524A90, 3211}, {"?", 0xFF7D77AF, 3241}, {"?", 0xFF9083A3, 3251}, {"?", 0xFF14214E, 3323}, {"?", 0xFF7F8BC2, 3331}, {"?", 0xFF1B3C78, 3333}, {"?", 0xFF2E4B9B, 3335}, {"?", 0xFF11263C, 3344}, {"?", 0xFF202A65, 3353}, {"?", 0xFF171B4A, 3355}, {"?", 0xFF002232, 3444}, {"?", 0xFF2D4491, 3522}, {"?", 0xFF261257, 3536}, {"?", 0xFF3A2885, 3541}, {"?", 0xFF233B7D, 3543}, {"?", 0xFF273C82, 3544}, {"?", 0xFF272651, 3554}, {"?", 0xFF28438C, 3600}, {"?", 0xFF243A7D, 3611}, {"?", 0xFF4055A1, 3612}, {"?", 0xFF1A4C8D, 3622}, {"?", 0xFF92B1DC, 3640}, {"?", 0xFF648DC7, 3641}, {"?", 0xFFD0DEEE, 3650}, {"?", 0xFFC8D6ED, 3652}, {"?", 0xFF00507F, 3732}, {"?", 0xFF12253C, 3743}, {"?", 0xFFB7D1E3, 3750}, {"?", 0xFFAFC9E5, 3761}, {"?", 0xFFCED2D1, 3770}, {"?", 0xFF3D6AA1, 3810}, {"?", 0xFF7BA2D3, 3815}, {"?", 0xFF91B9E2, 3820}, {"?", 0xFFB4CEEB, 3840}, {"?", 0xFF507193, 3842}, {"?", 0xFF007EBA, 3900}, {"?", 0xFF0082C4, 3901}, {"?", 0xFF006CA5, 3902}, {"?", 0xFF4ABDF0, 3910}, {"?", 0xFF86AACA, 3951}, {"?", 0xFF697698, 3953}, {"?", 0xFFA6D8F6, 3962}, {"?", 0xFFE1E1E1, 3971}, {"?", 0xFF0093B9, 4010}, {"?", 0xFF507793, 4032}, {"?", 0xFF265674, 4033}, {"?", 0xFFEAF0F9, 4071}, {"?", 0xFF838689, 4073}, {"?", 0xFF2DB0CF, 4101}, {"?", 0xFF0095C6, 4103}, {"?", 0xFF00A4D9, 4111}, {"?", 0xFF00A9C9, 4113}, {"?", 0xFF0082AD, 4116}, {"?", 0xFF00405D, 4133}, {"?", 0xFF192024, 4174}, / TODO: duplicate case value / **{"?", 0xFF4FB4CB, 4220}, {"?", 0xFF8DCEE4, 4230}, {"?", 0xFF8DCDDB, 4240}, {"?", 0xFFD5EBF2,**

4250}, {"?", 0xFF007B8D, 4410}, {"?", 0xFF0091A5, 4421}, {"?", 0xFF007D8C, 4423}, {"?", 0xFF007986, 4425}, {"?", 0xFF5FBFD1, 4430}, {"?", 0xFF006981, 4442}, {"?", 0xFF007F92, 4452}, {"?", 0xFF002F38, 4515}, {"?", 0xFF007389, 4531}, {"?", 0xFF007B8D, 4610}, / TODO: duplicate case value / **{"?", 0xFF00A3A0, 4620}, {"?", 0xFF0B7F85, 4625}, {"?", 0xFF005B63, 4643}, {"?", 0xFF234544, 4644}, {"?", 0xFF005B63, 5005},** / TODO: duplicate case value / **{"?", 0xFF00A6AD, 5010}, {"?", 0xFFB4DCD8, 5050}, {"?", 0xFF00876E, 5100}, {"?", 0xFF009084, 5101}, {"?", 0xFF48BAB7, 5115}, {"?", 0xFF00AF8C, 5210}, {"?", 0xFF8CCCC2, 5220}, {"?", 0xFF47B9AE, 5230}, {"?", 0xFF197E6D, 5233}, {"?", 0xFF006E42, 5324}, {"?", 0xFF004D3D, 5326}, {"?", 0xFF002F38, 5335},** / TODO: duplicate case value / **{"?", 0xFF002D1F, 5374}, {"?", 0xFF008663, 5411}, {"?", 0xFF006B4E, 5415}, {"?", 0xFF008663, 5422},** / TODO: duplicate case value / **{"?", 0xFF52BA84, 5500}, {"?", 0xFF14A363, 5510}, {"?", 0xFF007848, 5513}, {"?", 0xFF008663, 5515},** / TODO: duplicate case value */ {"?", 0xFF52A04F, 5531}, {"?", 0xFF94ADA1, 5552}, {"?", 0xFF103828, 5555}, {"?", 0xFF85C875, 5610}, {"?", 0xFF14B26D, 5613}, {"?", 0xFF1A781E, 5633}, {"?", 0xFF157436, 5643}, {"?", 0xFFC9E3C5, 5650}, {"?", 0xFF6B9181, 5664}, {"?", 0xFFA5C278, 5822}, {"?", 0xFF70953F, 5833}, {"?", 0xFF273014, 5866}, {"?", 0xFF81C750, 5912}, {"?", 0xFF457021, 5933}, {"?", 0xFF506702, 5934}, {"?", 0xFFBBDB41, 5940}, {"?", 0xFF003518, 5944}, {"?", 0xFFE3EB00, 6010}, {"?", 0xFFBED782, 6051}, {"?", 0xFF919600, 6133}, {"?", 0xFF484601, 6156}, {"END", 0, -1} };

Isafil Rayon Codes

thread_color Isafil_Rayon_codes[] = { {"?", 0xFFFFFFFF, 10}, {"?", 0xFFFFFFFF, 15}, /* TODO: duplicate case value / **{"?", 0xFFFFFFFF, 17},** / TODO: duplicate case value / **{"?", 0xFF000000, 20}, {"?", 0xFFFFFDED, 101}, {"?", 0xFF7D7D7D, 104}, {"?", 0xFF515B61, 107}, {"?", 0xFF6D757B, 108}, {"?", 0xFFACACAC, 109}, {"?", 0xFF515B61, 111},** / TODO: duplicate case value / **{"?", 0xFF5D5D5D, 112}, {"?", 0xFFCFCFCF, 124}, {"?", 0xFFA1A9B4, 131}, {"?", 0xFF6D757B, 141},** / TODO: duplicate case value / **{"?", 0xFF9EA5AA, 142}, {"?", 0xFFCFD1D5, 145}, {"?", 0xFFC6BDB4, 150}, {"?", 0xFFD5C4B3, 151}, {"?", 0xFF7C8283, 152}, {"?", 0xFF898F94, 156}, {"?", 0xFFFEF5F0, 180}, {"?", 0xFFE9D7D9, 182}, {"?", 0xFFEBE3DD, 184}, {"?", 0xFFE0DA5F, 221}, {"?", 0xFFBFBA28, 232}, {"?", 0xFFECE9C1, 241}, {"?", 0xFFFAF6CC, 250}, {"?", 0xFFECE7A5, 251}, {"?", 0xFFECEADB, 260}, {"?", 0xFFF9F8E8, 270}, {"?", 0xFFFDF76C, 310}, {"?", 0xFFF5D300, 311}, {"?", 0xFF797E24, 345}, {"?", 0xFFB0AA76, 352}, {"?", 0xFF898F2B, 442}, {"?", 0xFF98996D, 453}, {"?", 0xFF6E772E, 454}, {"?", 0xFF6B7E6F, 463}, {"?", 0xFF3E4F34, 465}, {"?", 0xFFEDEF05, 501}, {"?", 0xFFFAF6CC, 505},** / TODO: duplicate case value / **{"?", 0xFFF5D300, 506},** / TODO: duplicate case value / **{"?", 0xFFFFFBD1, 510}, {"?", 0xFFFDE896, 520}, {"?", 0xFFD7CE8A, 532}, {"?", 0xFFB18B00, 542}, {"?", 0xFFAA8D00, 545}, {"?", 0xFFB28F11, 546}, {"?", 0xFFAC9436, 551}, {"?", 0xFFB69F56, 552}, {"?", 0xFFF4EE8C, 580}, {"?", 0xFFF1EB35, 590}, {"?", 0xFFF8D73E, 600}, {"?", 0xFFF8D73E, 605},** / TODO: duplicate case value / **{"?", 0xFFF7CB47, 610}, {"?", 0xFFF7E400, 615}, {"?", 0xFFFDE896, 620},** / TODO: duplicate case value / **{"?", 0xFFEEDB00, 625}, {"?", 0xFFFEF09A, 630}, {"?", 0xFFFDE1AF, 635}, {"?", 0xFFFDE896, 640},** / TODO: duplicate case value / **{"?", 0xFFF5D2A6, 651}, {"?", 0xFFFEF9EA, 660}, {"?", 0xFFFAF6E7, 670}, {"?", 0xFFBEBEA8, 672}, {"?", 0xFFF7C35F, 700}, {"?", 0xFFF5CA00, 702}, {"?", 0xFFDFA200, 704}, {"?", 0xFFFCF538, 706}, {"?", 0xFFFADC59, 713}, {"?", 0xFF8C7E6A, 722}, {"?", 0xFF827000, 726}, {"?", 0xFF636254, 732}, {"?", 0xFF594900, 747}, {"?", 0xFFD6BF94, 761}, {"?", 0xFF656452, 776}, {"?", 0xFFF1AF00, 800}, {"?", 0xFFF3C200, 805}, {"?", 0xFFF5BA5D, 811}, {"?", 0xFFE1A23E, 821}, {"?", 0xFFCCAB3F, 822}, {"?", 0xFFDFA200, 824},** / TODO: duplicate case value / **{"?", 0xFFF3B044, 830}, {"?", 0xFFD0A44F, 832}, {"?", 0xFFCD944A, 842}, {"?", 0xFFE3BC61, 851}, {"?", 0xFF947C4A, 853}, {"?", 0xFF001F48, 866}, {"?", 0xFFEBE7DD, 870}, {"?", 0xFF9FA086, 873}, {"?", 0xFF9A897B,**

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JEF Colors

To do: find a citation

| Red | Green | Blue | Name | Catalog Number | |-----|-------|------|------|----------------| | 0 | 0 | 0 | Placeholder | 000 | | 0 | 0 | 0 | Black | 002 | | 255 | 255 | 255 | White | 001 | | 255 | 255 | 23 | Yellow | 204 | | 255 | 102 | 0 | Orange | 203 | | 47 | 89 | 51 | Olive Green | 219 | | 35 | 115 | 54 | Green | 226 | | 101 | 194 | 200 | Sky | 217 | | 171 | 90 | 150 | Purple | 208 | | 246 | 105 | 160 | Pink | 201 | | 255 | 0 | 0 | Red | 225 | | 177 | 112 | 78 | Brown | 214 | | 11 | 47 | 132 | Blue | 207 | | 228 | 195 | 93 | Gold | 003 | | 72 | 26 | 5 | Dark Brown | 205 | | 172 | 156 | 199 | Pale Violet | 209 | | 252 | 242 | 148 | Pale Yellow | 210 | | 249 | 153 | 183 | Pale Pink | 211 | | 250 | 179 | 129 | Peach | 212 | | 201 | 164 | 128 | Beige | 213 | | 151 | 5 | 51 | Wine Red | 215 | | 160 | 184 | 204 | Pale Sky | 216 | | 127 | 194 | 28 | Yellow Green | 218 | | 229 | 229 | 229 | Silver Gray | 220 | | 136 | 155 | 155 | Gray | 221 | | 152 | 214 | 189 | Pale Aqua | 227 | | 178 | 225 | 227 | Baby Blue | 228 | | 54 | 139 | 160 | Powder Blue | 229 | | 79 | 131 | 171 | Bright Blue | 230 | | 56 | 106 | 145 | Slate Blue | 231 | | 7 | 22 | 80 | Navy Blue | 232 | | 249 | 153 | 162 | Salmon Pink | 233 | | 249 | 103 | 107 | Coral | 234 | | 227 | 49 | 31 | Burnt Orange | 235 | | 226 | 161 | 136 | Cinnamon | 236 | | 181 | 148 | 116 | Umber | 237 | | 228 | 207 | 153 | Blond | 238 | | 255 | 203 | 0 | Sunflower | 239 | | 225 | 173 | 212 | Orchid Pink | 240 | | 195 | 0 | 126 | Peony Purple | 241 | | 128 | 0 | 75 | Burgundy | 242 | | 84 | 5 | 113 | Royal Purple | 243 | | 177 | 5 | 37 | Cardinal Red | 244 | | 202 | 224 | 192 | Opal Green | 245 | | 137 | 152 | 86 | Moss Green | 246 | | 92 | 148 | 26 | Meadow Green | 247 | | 0 | 49 | 20 | Dark Green | 248 | | 93 | 174 | 148 | Aquamarine | 249 | | 76 | 191 | 143 | Emerald Green | 250 | | 0 | 119 | 114 | Peacock Green | 251 | | 89 | 91 | 97 | Dark Gray | 252 | | 255 | 255 | 242 | Ivory White | 253 | | 177 | 88 | 24 | Hazel | 254 | | 203 | 138 | 7 | Toast | 255 | | 152 | 108 | 128 | Salmon | 256 | | 152 | 105 | 45 | Cocoa Brown | 257 | | 77 | 52 | 25 | Sienna | 258 | | 76 | 51 | 11 | Sepia | 259 | | 51 | 32 | 10 | Dark Sepia | 260 | | 82 | 58 | 151 | Violet Blue | 261 | | 13 | 33 | 126 | Blue Ink | 262 | | 30 | 119 | 172 | Sola Blue | 263 | | 178 | 221 | 83 | Green Dust | 264 | | 243 | 54 | 137 | Crimson | 265 | | 222 | 100 | 158 | Floral Pink | 266 | | 152 | 65 | 97 | Wine | 267 | | 76 | 86 | 18 | Olive Drab | 268 | | 76 | 136 | 31 | Meadow | 269 | | 228 | 222 | 121 | Mustard | 270 | | 203 | 138 | 26 | Yellow Ocher | 271 | | 203 | 162 | 28 | Old Gold | 272 | | 255 | 152 | 5 | Honey Dew | 273 | | 252 | 178 | 87 | Tangerine | 274 | | 0xFF | 0xE5 | 0x05 | Canary Yellow | 275 | 0xF0 | 0x33 | 0x1F | Vermilion | 202 | | 0x1A | 0x84 | 0x2D | Bright Green | 206 | | 0x38 | 0x6C | 0xAE | Ocean Blue | 222 | | 0xE3 | 0xC4 | 0xB4 | Beige Gray | 223 | | 0xE3 | 0xAC | 0x81 | Bamboo | 224 |

Madeira Polyester Codes

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Madeira Rayon Codes

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Marathon Polyester Codes

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Marathon Rayon Codes

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PCM Color Codes

;const EmbThread pcmThreads[] = { 0x00, 0x00, 0x00, "PCM Color 1", "PCM Color 1" 0x00, 0x00, 0x80, "PCM Color 2", "PCM Color 2" 0x00, 0x00, 0xFF, "PCM Color 3", "PCM Color 3" 0x00, 0x80, 0x80, "PCM Color 4", "PCM Color 4" 0x00, 0xFF, 0xFF, "PCM Color 5", "PCM Color 5" 0x80, 0x00, 0x80, "PCM Color 6", "" 0xFF, 0x00, 0xFF, "PCM Color 7", "" 0x80, 0x00, 0x00, "PCM Color 8", "" 0xFF, 0x00, 0x00, "PCM Color 9", "" 0x00, 0x80, 0x00, "PCM Color 10", "" 0x00, 0xFF, 0x00, "PCM Color 11", "" 0x80, 0x80, 0x00, "PCM Color 12", "" 0xFF, 0xFF, 0x00, "PCM Color 13", "" 0x80, 0x80, 0x80, "PCM Color 14", "" 0xC0, 0xC0, 0xC0, "PCM Color 15", "" 0xFF, 0xFF, 0xFF, "PCM Color 16", "" 0, 0, 0, "END", "END"

PEC Color Codes

; ; 0, 0, 0, "Unknown", "index 0" 14, 31, 124, "Prussian Blue", "index 1" 10, 85, 163, "Blue", "" ; Index 2 0, 135, 119, "Teal Green", "" ; Index 3 75, 107, 175, "Cornflower Blue", "" ; Index 4 237, 23, 31, "Red", "" ; Index 5 209, 92, 0, "Reddish Brown", "" ; Index 6 145, 54, 151, "Magenta", "" ; Index 7 228, 154, 203, "Light Lilac", "" ; Index 8 145, 95, 172, "Lilac", "" ; Index 9 158, 214, 125, "Mint Green", "" ; Index 10 232, 169, 0, "Deep Gold", "" ; Index 11 254, 186, 53, "Orange", "" ; Index 12 255, 255, 0, "Yellow", "" ; Index 13 112, 188, 31, "Lime Green", "" ; Index 14 186, 152, 0, "Brass", "" ; Index 15 168, 168, 168, "Silver", "" ; Index 16 ; TODO: Verify RGB value is correct 125, 111, 0, "Russet Brown", "" ; Index 17 255, 255, 179, "Cream Brown", "" ; Index 18 79, 85, 86, "Pewter", "" ; Index 19 0, 0, 0, "Black", "" ; Index 20 11, 61, 145, "Ultramarine", "" ; Index 21 119, 1, 118, "Royal Purple", "" ; Index 22 41, 49, 51, "Dark Gray", "" ; Index 23 42, 19, 1, "Dark Brown", "" ; Index 24 246, 74, 138, "Deep Rose", "" ; Index 25 178, 118, 36, "Light Brown", "" ; Index 26 ; TODO: Verify RGB value is correct 252, 187, 197, "Salmon Pink", "" ; Index 27 254, 55, 15, "Vermillion", "" ; Index 28 240, 240, 240, "White", "" ; Index 29 106, 28, 138, "Violet", "" ; Index 30 168, 221, 196, "Seacrest", "" ; Index 31 37, 132, 187, "Sky Blue", "" ; Index 32 254, 179, 67, "Pumpkin", "" ; Index 33 255, 243, 107, "Cream Yellow", "" ; Index 34 208, 166, 96, "Khaki", "" ; Index 35 209, 84, 0, "Clay Brown", "" ; Index 36 102, 186, 73, "Leaf Green", "" ; Index 37 19, 74, 70, "Peacock Blue", "" ; Index 38 135, 135, 135, "Gray", "" ; Index 39 ; TODO: Verify RGB value is correct 216, 204, 198, "Warm Gray", "" ; Index 40 67, 86, 7, "Dark Olive", "" ; Index 41 ; TODO: Verify RGB value is correct 253, 217, 222, "Flesh Pink", "" ; Index 42 249, 147, 188, "Pink", "" ; Index 43 0, 56, 34, "Deep Green", "" ; Index 44 178, 175, 212, "Lavender", "" ; Index 45 104, 106, 176, "Wisteria Violet", "" ; Index 46 239, 227, 185, "Beige", "" ; Index 47 247, 56, 102, "Carmine", "" ; Index 48 ; TODO: Verify RGB value is correct 181, 75, 100, "Amber Red", "" ; Index 49 19, 43, 26, "Olive Green", "" ; Index 50 ; TODO: Verify RGB value is correct 199, 1, 86, "Dark Fuschia", "" ; Index 51 254, 158, 50, "Tangerine", "" ; Index 52 168, 222, 235, "Light Blue", "" ; Index 53 ; TODO: Verify RGB value is correct 0, 103, 62, "Emerald Green", "" ; Index 54 78, 41, 144, "Purple", "" ; Index 55 47, 126, 32, "Moss Green", "" ; Index

56 ; TODO: Verify RGB value is correct ; TODO: Flesh Pink is Index 42, is this Index incorrect? 255, 204, 204, "Flesh Pink", "" ; Index 57 255, 217, 17, "Harvest Gold", "" ; Index 58 9, 91, 166, "Electric Blue", "" ; Index 59 240, 249, 112, "Lemon Yellow", "" ; Index 60 227, 243, 91, "Fresh Green", "" ; Index 61 ; TODO: Verify RGB value is correct ; TODO: Orange is Index 12, is this Index incorrect? 255, 153, 0, "Orange", "" ; Index 62 ; TODO: Verify RGB value is correct ; TODO: Cream Yellow is Index 34, is this Index incorrect? 255, 240, 141, "Cream Yellow", "" ; Index 63 255, 200, 200, "Applique", "" ; Index 64 0, 0, 0, "END", "END"

Robinson Anton Polyester Codes

thread_color RobisonAnton_Polyester_codes[] = { {"Black", 0xFF000000, 5596}, {"White", 0xFFFFFFFF, 5597}, /* unfinished */ {"END", 0, -1} };

thread_color RobisonAnton_Rayon_codes[] = { {"Light Pink", 0xFFEFCCCE, 2243}, {"Pink", 0xFFFCBFC9, 2223}, {"Pink Bazaar", 0xFFEFC6D3, 2599}, {"Pink Mist", 0xFFF9B2B7, 2373}, {"Emily Pink", 0xFFF9AFAD, 2374}, {"Rose", 0xFFFC9BB2, 2293}, {"Rose Cerise", 0xFFFC8C99, 2244}, {"Carnation", 0xFFF2AFC1, 2237}, {"Shrimp", 0xFFE5566D, 2246}, {"Bashful Pink", 0xFFF4476B, 2248}, {"Begonia", 0xFFE5566D, 2228}, /* TODO: duplicate case value / **{"Azalea", 0xFFF9848E, 2412}, {"Dusty Rose", 0xFFF26877, 2375}, {"Rose Tint", 0xFFD8899B, 2591}, {"Burgundy", 0xFF8C2633, 2249}, {"TH Burgundy", 0xFF7C2128, 2608}, {"Russet", 0xFF7A2638, 2252}, {"Pro Firebrand", 0xFF75263D, 2622}, {"Wine", 0xFF772D35, 2225}, {"Intense Maroon", 0xFF593344, 2587}, {"Dark Maroon", 0xFF4F213A, 2376}, {"Carbernet", 0xFF931638, 2494}, {"Mountain Rose", 0xFFF9B2B7, 2495},** / TODO: duplicate case value / **{"Warm Wine", 0xFF661E2B, 2496}, {"Primrose", 0xFFE5566D, 2491},** / TODO: duplicate case value / **{"Perfect Ruby", 0xFF8E2344, 2497}, {"Brushed Burgundy", 0xFF6D213F, 2498}, {"Passion Rose", 0xFF8E2344, 2499},** / TODO: duplicate case value / **{"New Berry", 0xFFAD0075, 2500}, {"Petal Pink", 0xFFFCC9C6, 2501}, {"Memphis Belle", 0xFFF4BFD1, 2502}, {"Desert Bloom", 0xFFF7BFBF, 2503}, {"Wild Pink", 0xFFCE007C, 2259}, {"Floral Pink", 0xFFED72AA, 2415}, {"Hot Pink", 0xFFD36B9E, 2260}, {"Crimson", 0xFFD60270, 2416}, {"Ruby Glint", 0xFFE22882, 2261}, {"Cherrystone", 0xFFAA004F, 2504}, {"Cherry Punch", 0xFFAA004F, 2417},** / TODO: duplicate case value / **{"Cherry Blossom", 0xFFEA0F6B, 2262}, {"Red Berry", 0xFFAF1E2D, 2418}, {"Jockey Red", 0xFFBF0A30, 2281}, {"Very Red", 0xFFBF0A30, 2419},** / TODO: duplicate case value / **{"Red Berry", 0xFFBF0A30, 2378},** / TODO: duplicate case value / **{"Foxy Red", 0xFFCE1126, 2263}, {"Tuxedo Red", 0xFFD62828, 2420}, {"Lipstick", 0xFFBF0A30, 2233},** / TODO: duplicate case value / **{"Scarlet", 0xFFC41E3A, 2219}, {"Radiant Red", 0xFFC41E3A, 2266},** / TODO: duplicate case value / **{"Wildfire", 0xFFA32638, 2267}, {"Carolina Red", 0xFF8C2633, 2268},** / TODO: duplicate case value / **{"Red Jubilee", 0xFF75263D, 2421},** / TODO: duplicate case value / **{"Cranberry", 0xFF992135, 2270}, {"Antique Red", 0xFFA32638, 2505},** / TODO: duplicate case value / **{"Devil Red", 0xFFC41E3A, 2506},** / TODO: duplicate case value / **{"Rosewood", 0xFFD81C3F, 2508}, {"Bitteroot", 0xFFFC5E72, 2509}, {"Bisque", 0xFFF2C4AF, 2377}, {"Flesh", 0xFFF4CCAA, 2413}, {"Flesh Pink", 0xFFF9BAAA, 2253}, {"Opal Mist", 0xFFF7BFBF, 2255},** / TODO: duplicate case value / **{"Candy Apple Red", 0xFFAF003D, 2507}, {"Tawny", 0xFFF9BF9E, 2256}, {"Peach", 0xFFF9C6AA, 2257}, {"Melon", 0xFFF98E6D, 2294}, {"Flamingo", 0xFFF9A58C, 2258}, {"Coral", 0xFFF98972, 2414}, {"Persimmon", 0xFFF43F4F, 2277}, {"Peach Blossom", 0xFFF9C6AA, 2510},** / TODO: duplicate case value / **{"Illusion", 0xFFF9BAAA, 2511},** / TODO: duplicate case value / **{"Melonade", 0xFFF98972, 2512},** / TODO: duplicate case value / **{"Honeysuckle", 0xFFE23D28, 2513}, {"Brite Jade", 0xFF008C82, 2514}, {"Bluestone", 0xFF00B2A0, 2515}, {"Aqua Pearl",**

0xFF47D6C1, 2516}, {"Seafrost", 0xFF87DDD1, 2517}, {"J. Turquoise", 0xFF008789, 2492}, {"Indian Ocean Blue", 0xFF2DC6D6, 2518}, {"Surf Blue", 0xFF00A5DB, 2519}, {"Mid Windsor", 0xFF00A3DD, 2520}, {"Deep Windsor", 0xFF003F54, 2589}, {"Pro Dark Blue", 0xFF002D47, 2620}, {"Mallard Blue", 0xFF006D75, 2521}, {"Sky Blue", 0xFF9BC4E2, 2239}, {"Lake Blue", 0xFFAFBCDB, 2304}, {"Pro Lusty Blue", 0xFF5B77CC, 2614}, {"Slate Blue", 0xFF6D87A8, 2275}, {"Blue Frost", 0xFF99D6DD, 2305}, {"Periwinkle", 0xFF28C4D8, 2306}, {"Aquamarine", 0xFF00ADC6, 2307}, {"California Blue", 0xFF00A0C4, 2389}, {"Baltic Blue", 0xFF008ED6, 2441}, {"Solar Blue", 0xFF0054A0, 2442}, {"Pacific Blue", 0xFF008ED6, 2388}, / TODO: duplicate case value / **{"Boo Boo Blue", 0xFF00A3DD, 2730},** / TODO: duplicate case value / **{"Pro Band Blue", 0xFF00709E, 2737}, {"Pro Peacock", 0xFF007AA5, 2740}, {"Light Blue", 0xFFC9E8DD, 2222}, {"Royal", 0xFF1E1C77, 2210}, {"Blue Suede", 0xFF002B7F, 2438}, {"Imperial Blue", 0xFF335687, 2302}, {"Bridgeport Blue", 0xFFAFBCDB, 2522},** / TODO: duplicate case value / **{"China Blue", 0xFF335687, 2523},** / TODO: duplicate case value / **{"Pro Imperial", 0xFF26547C, 2612}, {"Country Blue", 0xFF7796B2, 2524}, {"Heron Blue", 0xFFC1C9DD, 2525}, {"Pro Saxon", 0xFF7796B2, 2624},** / TODO: duplicate case value / **{"Bright Blue", 0xFF6689CC, 2526}, {"Soldier Blue", 0xFF5960A8, 2527}, {"Pro Brilliance", 0xFF0051BA, 2619}, {"Atlantis Blue", 0xFF6689CC, 2528},** / TODO: duplicate case value / **{"Dolphin Blue", 0xFF3A75C4, 2529}, {"Caribbean Blue", 0xFF75AADB, 2530}, {"Dana Blue", 0xFFC4D8E2, 2531}, {"Cadet Blue", 0xFFC4D8E2, 2532},** / TODO: duplicate case value / **{"Ozone", 0xFF60AFDD, 2533}, {"Salem Blue", 0xFF003D6B, 2534}, {"Blue Ribbon", 0xFF0F2B5B, 2439}, {"Pro Navy", 0xFF192168, 2625}, {"Blue Ink", 0xFF2B265B, 2440}, {"Pro Midnight", 0xFF002654, 2613}, {"Pro College Blue", 0xFF002649, 2647}, {"Light Midnight", 0xFF353F5B, 2386}, {"Fleet Blue", 0xFF35264F, 2450}, {"Light Navy", 0xFF112151, 2303}, {"Flag Blue", 0xFF14213D, 2603}, {"TH Navy", 0xFF002654, 2609},** / TODO: duplicate case value / **{"Navy", 0xFF14213D, 2215},** / TODO: duplicate case value / **{"Midnight Navy", 0xFF14213D, 2387},** / TODO: duplicate case value / **{"Pastel Blue", 0xFFC6D1D6, 2382}, {"Blue Hint", 0xFFC1C9DD, 2598},** / TODO: duplicate case value / **{"Ice Blue", 0xFFB5D1E8, 2300}, {"Paris Blue", 0xFFD1CEDD, 2283}, {"Baby Blue", 0xFF99BADD, 2206}, {"Sun Blue", 0xFFAFBCDB, 2269},** / TODO: duplicate case value / **{"Cristy Blue", 0xFFA5BAE0, 2383}, {"Ultra Blue", 0xFF75B2DD, 2433}, {"Tropic Blue", 0xFF75AADB, 2434},** / TODO: duplicate case value / **{"Blue Horizon", 0xFF6689CC, 2435},** / TODO: duplicate case value / **{"Oriental Blue", 0xFF7F8CBF, 2301}, {"Copen", 0xFF6D87A8, 2245},** / TODO: duplicate case value / **{"Jay Blue", 0xFF5B77CC, 2384},** / TODO: duplicate case value / **{"Blue", 0xFF2D338E, 2220}, {"Sapphire", 0xFF2D338E, 2280},** / TODO: duplicate case value / **{"Pro Royal", 0xFF00337F, 2627}, {"Fire Blue", 0xFF003893, 2436}, {"Jamie Blue", 0xFF2D338E, 2385},** / TODO: duplicate case value / **{"Empire Blue", 0xFF3F2893, 2437}, {"Enchanted Sea", 0xFF3A564F, 2535}, {"Pro Twinkle", 0xFF8499A5, 2617}, {"Rockport Blue", 0xFF9BAABF, 2536}, {"Wonder Blue", 0xFF5E99AA, 2577}, {"Traditional Gray", 0xFFADAFAA, 2540}, {"Steel", 0xFFBFBAAF, 2537}, {"Pro Pearl", 0xFFBAB7AF, 2741}, {"Pro Cool Gray", 0xFFC4C1BA, 2733}, {"Stainless Steel", 0xFFCCC1B2, 2538}, {"Chrome", 0xFFD1CCBF, 2539}, {"Pro Night Sky", 0xFFADAFAA, 2618},** / TODO: duplicate case value / **{"Gull", 0xFF8C8984, 2731}, {"Mineral", 0xFF686663, 2729}, {"Black Chrome", 0xFF443D38, 2541}, {"Heather", 0xFFDDC6C4, 2271}, {"Grape", 0xFFD3B7A3, 2272}, {"Satin Wine", 0xFFB5939B, 2314}, {"Ducky Mauve", 0xFF8E6877, 2422}, {"Pale Orchid", 0xFFFCBFC9, 2423},** / TODO: duplicate case value / **{"Orchid", 0xFFE5BFC6, 2379}, {"Lavender", 0xFFEFC6D3, 2276},** / TODO: duplicate case value / **{"Violet", 0xFFEDC4DD, 2285}, {"Cachet", 0xFFB58CB2, 2424}, {"Tulip", 0xFFC9ADD8, 2286}, {"Mid Lilac", 0xFFE29ED6, 2588}, {"Port Wine", 0xFF512654, 2600}, {"Pro Maroon", 0xFF512D44, 2616}, {"Laurie Lilac", 0xFF8E47AD, 2425}, {"Iris", 0xFFAF72C1, 2288}, {"Raspberry", 0xFF9B4F96, 2426}, {"Mulberry", 0xFF66116D, 2380}, {"Plum Wine", 0xFF63305E, 2490}, {"Purple Twist", 0xFF1E1C77, 2429},** / TODO: duplicate case value / **{"Violet Blue",**

0xFF332875, 2427}, {"Purple Maze", 0xFF35006D, 2428}, {"Pro Brite Star", 0xFF2B1166, 2736}, {"Pro Violet", 0xFF38197A, 2742}, {"Pro Purple", 0xFF35006D, 2628}, / TODO: duplicate case value / **{"Purple", 0xFF8E47AD, 2254},** / TODO: duplicate case value / **{"Purple Shadow", 0xFF5B027A, 2430}, {"Dark Purple", 0xFF4C145E, 2381}, {"Mauve", 0xFF8977BA, 2287}, {"Purple Accent", 0xFF44235E, 2431}, {"Hot Peony", 0xFFAA0066, 2590}, {"Passion", 0xFFCE007C, 2291},** / TODO: duplicate case value / **{"Strawberry", 0xFFAA004F, 2432},** / TODO: duplicate case value / **{"Plum", 0xFF9E2387, 2292}, {"Misty", 0xFF5E99AA, 2308},** / TODO: duplicate case value / **{"Mystic Teal", 0xFF609191, 2443}, {"Teal", 0xFF609191, 2309},** / TODO: duplicate case value / **{"Dark Teal", 0xFF003F54, 2444},** / TODO: duplicate case value / **{"Mint Julep", 0xFF93DDDB, 2310}, {"Turquoise", 0xFF7FD6DB, 2204}, {"M.D. Green", 0xFF007272, 2445}, {"Seafoam", 0xFF70CE9B, 2311}, {"Isle Green", 0xFF70CE9B, 2312},** / TODO: duplicate case value / **{"Peppermint", 0xFF35C4AF, 2390}, {"Oceanic Green", 0xFF006D75, 2446},** / TODO: duplicate case value / **{"Pro Teal", 0xFF006B77, 2621}, {"Garden Green", 0xFF006663, 2447}, {"Pro Green", 0xFF006D66, 2735}, {"Pine Green", 0xFF008C82, 2391},** / TODO: duplicate case value / **{"Greenstone", 0xFF008272, 2448}, {"Fern Green", 0xFF006663, 2449},** / TODO: duplicate case value / **{"Pro Hunter", 0xFF006D66, 2615},** / TODO: duplicate case value / **{"Palm Leaf", 0xFFBCC1B2, 2241}, {"Flite Green", 0xFFC6D6A0, 2282}, {"Willow", 0xFF9EAA99, 2221}, {"Sprite", 0xFFB2D8D8, 2313}, {"Moss", 0xFF7AA891, 2278}, {"Wintergreen", 0xFF7AA891, 2594},** / TODO: duplicate case value / **{"Green Forest", 0xFF006056, 2451}, {"Pro Forest", 0xFF006056, 2743},** / TODO: duplicate case value / **{"Harbor Green", 0xFF4F6D5E, 2392}, {"Evergreen", 0xFF024930, 2315}, {"Pro Dark Green", 0xFF004438, 2734}, {"Lizzy Lime", 0xFF2B4C3F, 2631}, {"D.H. Green", 0xFF282D26, 2411}, {"Celery", 0xFFF2EABC, 2316}, {"Pistachio", 0xFFCCC693, 2250}, {"Olive Drab", 0xFF5E663A, 2317}, {"Olive", 0xFF779182, 2202}, {"Pale Green", 0xFFC9E8DD, 2318},** / TODO: duplicate case value / **{"Green Pearl", 0xFF93DDDB, 2452},** / TODO: duplicate case value / **{"Sea Mist", 0xFF93DDDB, 2393},** / TODO: duplicate case value / **{"Mint", 0xFFB5E8BF, 2238}, {"Spruce", 0xFFAADD6D, 2279}, {"Nile", 0xFFA0DB8E, 2211}, {"Green Oak", 0xFFB5CC8E, 2319}, {"Erin Green", 0xFF8CD600, 2320}, {"Pro Erin", 0xFF56AA1C, 2738}, {"Emerald", 0xFF339E35, 2214}, {"Dark Emerald", 0xFF007A3D, 2453}, {"Light Kelly", 0xFF007A3D, 2410},** / TODO: duplicate case value / **{"Kelly", 0xFF007A3D, 2240},** / TODO: duplicate case value / **{"Dark Green", 0xFF008751, 2208}, {"Fleece Green", 0xFF006B3F, 2454}, {"TH Green", 0xFF006854, 2607}, {"Harvest Green", 0xFF1EB53A, 2578}, {"Vibrant Green", 0xFF009E49, 2579}, {"Green Grass", 0xFF009E49, 2580},** / TODO: duplicate case value / **{"Deep Green", 0xFF006B3F, 2284},** / TODO: duplicate case value / **{"Green Bay", 0xFF006B54, 2455}, {"Jungle Green", 0xFF007C66, 2597}, {"Peapod", 0xFFA3AF07, 2456}, {"Pastoral Green", 0xFF7FBA00, 2321}, {"Green Dust", 0xFF7FBA00, 2457},** / TODO: duplicate case value / **{"Ming", 0xFF7FBA00, 2322},** / TODO: duplicate case value / **{"Meadow", 0xFF568E14, 2226}, {"Tamarack", 0xFF939905, 2230}, {"Palmetto", 0xFF566314, 2229}, {"Green Petal", 0xFF024930, 2458},** / TODO: duplicate case value / **{"Sage", 0xFF547730, 2595}, {"Hedge", 0xFF3F4926, 2601}, {"Green", 0xFF3A7728, 2209}, {"Green Sail", 0xFF193833, 2459}, {"Holly", 0xFF215B33, 2323}, {"Field Green", 0xFF265142, 2460}, {"Dress Green", 0xFF3F4926, 2584},** / TODO: duplicate case value / **{"Foliage Green", 0xFF99840A, 2542}, {"Autumn Green", 0xFFA38205, 2543}, {"Desert Cactus", 0xFF897719, 2544}, {"Cypress", 0xFF707014, 2545}, {"Crescent Moon", 0xFF848205, 2546}, {"Pebblestone", 0xFFE2E584, 2547}, {"Sun Shadow", 0xFF998E07, 2548}, {"Blue Spruce", 0xFF00494F, 2549}, {"Newport", 0xFF4F6D5E, 2550},** / TODO: duplicate case value / **{"Spring Garden", 0xFF779182, 2551},** / TODO: duplicate case value / **{"Water Lilly", 0xFF546856, 2554}, {"Ivy", 0xFF0C3026, 2552}, {"Dark Army Green", 0xFF233A2D, 2553}, {"Army Green", 0xFF213D30, 2728}, {"Pastel Green", 0xFFC9D6A3, 2555}, {"Pollen Gold", 0xFFE0AA0F, 2556}, {"Pale Yellow", 0xFFF4E287, 2557}, {"Buttercup",**

0xFFFFC61E, 2558}, {"Tusk", 0xFFF7E8AA, 2559}, {"Moonbeam", 0xFFF9DD16, 2560}, {"Black Eyed Susie", 0xFFC6A00C, 2561}, {"Bullion", 0xFFA37F14, 2562}, {"Chinese Yellow", 0xFFF7E8AA, 2324}, / TODO: duplicate case value / **{"Maize", 0xFFF7E8AA, 2264},** / TODO: duplicate case value / **{"Wheat", 0xFFEADD96, 2461}, {"Pro Maize", 0xFFF9E08C, 2732}, {"Glow", 0xFFFFD87F, 2234}, {"Star Gold", 0xFFFCD856, 2408}, {"Mango", 0xFFFCA311, 2394}, {"Yellow Mist", 0xFFFCA311, 2409},** / TODO: duplicate case value / **{"Yellow", 0xFFFCA311, 2213},** / TODO: duplicate case value / **{"Sunflower", 0xFFF9DD16, 2462},** / TODO: duplicate case value / **{"Lemon", 0xFFF4ED47, 2325}, {"Daffodil", 0xFFF9E814, 2326}, {"Merit Gold", 0xFFFCB514, 2463}, {"Cornsilk", 0xFFFFCC49, 2395}, {"Nectar", 0xFFFFC61E, 2464},** / TODO: duplicate case value / **{"Scholastic", 0xFFFCBF49, 2465}, {"Canary Yellow", 0xFFFCE016, 2235}, {"Pro Gold", 0xFFFFC61E, 2626},** / TODO: duplicate case value / **{"Manila", 0xFFFCD116, 2466}, {"Goldenrod", 0xFFFCD116, 2242},** / TODO: duplicate case value / **{"Brite Yellow", 0xFFFFCC49, 2396},** / TODO: duplicate case value / **{"Honeydew", 0xFFF99B0C, 2327}, {"Pumpkin", 0xFFF77F00, 2328}, {"Orangeade", 0xFFF74902, 2467}, {"Sun Orange", 0xFFF74902, 2397},** / TODO: duplicate case value / **{"Paprika", 0xFFF95602, 2236}, {"Saffron", 0xFFF93F26, 2329}, {"Tex Orange", 0xFFF95602, 2468},** / TODO: duplicate case value / **{"Orange", 0xFFF96B07, 2218}, {"Dark Tex Orange", 0xFFF96302, 2469}, {"Old Dark Tex Orange", 0xFFA53F0F, 2581}, {"Golden Poppy", 0xFFFC8744, 2330}, {"Rust", 0xFFBC4F07, 2289}, {"Copper", 0xFFAF7505, 2295}, {"Light Bronze", 0xFFC18E60, 2493}, {"Visor Gold", 0xFFFCCE87, 2398}, {"Goldenlite", 0xFFFFCC49, 2605},** / TODO: duplicate case value / **{"Honey", 0xFFFCBA5E, 2247}, {"Marigold", 0xFFFFCC49, 2216},** / TODO: duplicate case value / **{"Mustard", 0xFFE0AA0F, 2331},** / TODO: duplicate case value / **{"Sun Gold", 0xFFE0AA0F, 2212},** / TODO: duplicate case value / **{"Karat", 0xFFE0AA0F, 2470},** / TODO: duplicate case value / **{"Penny", 0xFFF2BF49, 2332}, {"New Gold", 0xFFFCCE87, 2399},** / TODO: duplicate case value / **{"Pro Beige", 0xFFE2D6B5, 2630}, {"Marine Gold", 0xFFC6A00C, 2596},** / TODO: duplicate case value / **{"Ginger", 0xFFA37F14, 2333},** / TODO: duplicate case value / **{"Shimmering Gold", 0xFFC6A00C, 2471},** / TODO: duplicate case value / **{"Old Gold", 0xFFBF910C, 2201}, {"Salmon", 0xFFEAB2B2, 2299}, {"Dark Rust", 0xFFC13828, 2205}, {"Terra Cotta", 0xFFA03033, 2334}, {"Pro Red", 0xFFAF1E2D, 2623},** / TODO: duplicate case value / **{"Auburn", 0xFFAF1E2D, 2472},** / TODO: duplicate case value / **{"Bone", 0xFFFAE6CC, 2582}, {"Ivory", 0xFFFAE6CF, 2335}, {"Opaline", 0xFFF7D3B5, 2473}, {"Ecru", 0xFFEDD3BC, 2232}, {"Wicker", 0xFFAA753F, 2489}, {"Tan", 0xFFD3A87C, 2273}, {"Cottage Beige", 0xFFEDD3B5, 2593}, {"Rattan", 0xFFC1A875, 2474}, {"Gold", 0xFFE2BF9B, 2203}, {"Mocha Cream", 0xFFD3A87C, 2475},** / TODO: duplicate case value / **{"Topaz", 0xFFBF910C, 2400},** / TODO: duplicate case value / **{"Ashley Gold", 0xFFD18E54, 2401}, {"Amber Beige", 0xFFD8B596, 2336}, {"Seashell", 0xFFD6CCAF, 2476}, {"Light Maize", 0xFFF2E3C4, 2604}, {"Beige", 0xFFAA753F, 2224},** / TODO: duplicate case value / **{"Sand Dune", 0xFF6B4714, 2477}, {"Taupe", 0xFFD1BF91, 2298}, {"Pro Brown", 0xFF6C463D, 2610}, {"Chocolate", 0xFF876028, 2227}, {"Pro Walnut", 0xFF755426, 2629}, {"Light Cocoa", 0xFF755426, 2478},** / TODO: duplicate case value / **{"Cocoa Mulch", 0xFFB28260, 2488}, {"Brown", 0xFF593D2B, 2251}, {"Dark Brown", 0xFF593D2B, 2372},** / TODO: duplicate case value / **{"Espresso", 0xFF3F302B, 2337}, {"Bamboo", 0xFFC18E60, 2338},** / TODO: duplicate case value / **{"Almond", 0xFFAF7505, 2479},** / TODO: duplicate case value / **{"Toast", 0xFFBA7530, 2231}, {"Sienna", 0xFF755426, 2402},** / TODO: duplicate case value / **{"K.A. Bronze", 0xFFB26B70, 2480}, {"Pro Cinnamon", 0xFFA2464E, 2611}, {"Date", 0xFF9B4F19, 2290}, {"Hazel", 0xFF9B4F19, 2481},** / TODO: duplicate case value / **{"Coffee Bean", 0xFF5B2D28, 2339}, {"Dogwood", 0xFF593D2B, 2563},** / TODO: duplicate case value / **{"Mahogany", 0xFF3F302B, 2564},** / TODO: duplicate case value / **{"Best Brown", 0xFF3D3028, 2566}, {"Mushroom", 0xFF633A11, 2567}, {"Perfect Tan", 0xFFC1A875, 2568},** / TODO: duplicate case value / **{"Earthen Tan", 0xFF7A5B11, 2569}, {"Golden Tan", 0xFFC1A875, 2570},** / TODO: duplicate case value / **{"14**

Kt. Gold", 0xFFF2BF49, 2586}, / TODO: duplicate case value / **{"TH Gold", 0xFFF2CE68, 2606}, {"24 Kt. Gold", 0xFFD88C02, 2602}, {"Platinum", 0xFFC1B5A5, 2571}, {"Pro Gray", 0xFF99897C, 2739}, {"Grayrod", 0xFFADA07A, 2572}, {"Pewter", 0xFFADA07A, 2573},** / TODO: duplicate case value / **{"Aspen White", 0xFFF5E3CC, 2574}, {"Dark Taupe", 0xFF66594C, 2575}, {"Egyptian Brown", 0xFF493533, 2576}, {"Oyster", 0xFFF5EBE0, 2403}, {"Gray", 0xFFDDC6C4, 2207},** / TODO: duplicate case value / **{"Pearl Gray", 0xFFDBD3D3, 2340}, {"Steel Gray", 0xFFD8CCD1, 2274}, {"Skylight", 0xFFCCC1C6, 2482}, {"Cloud", 0xFFAFAAA3, 2483}, {"Silver Steel", 0xFFADAFAA, 2592},** / TODO: duplicate case value / **{"Banner Gray", 0xFF919693, 2585}, {"Silvery Gray", 0xFF8C8984, 2484},** / TODO: duplicate case value / **{"Cinder", 0xFFCCC1C6, 2404},** / TODO: duplicate case value / **{"Saturn Gray", 0xFFDBD3D3, 2485},** / TODO: duplicate case value / **{"Dover Gray", 0xFFCCC1C6, 2405},** / TODO: duplicate case value / **{"Storm Gray", 0xFFB2A8B5, 2486}, {"Sterling", 0xFFA893AD, 2406}, {"Metal", 0xFF666D70, 2407}, {"Twilight", 0xFF686663, 2217},** / TODO: duplicate case value / **{"Aged Charcoal", 0xFF443D38, 2565},** / TODO: duplicate case value / **{"Charcoal", 0xFF777772, 2265}, {"Smokey", 0xFF353842, 2487}, {"Ash", 0xFF3A4972, 2341}, {"Black", 0xFF1C2630, 2296}, {"Snow White", 0xFFF5EBE0, 2297},** / TODO: duplicate case value / **{"Natural White", 0xFFF5EDDE, 2342}, {"Eggshell", 0xFFF0E8D6, 2343}, {"Jet Black", 0xFF1C2630, 2632},** / TODO: duplicate case value */ {"END", 0, -1} };

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Sigma Polyester Codes

"","Embroidermodder 2 CSV Embroidery File" "","http://embroidermodder.github.io" "","" "","General Notes:" "","This file can be read by Excel or LibreOffice as CSV (Comma Separated Value) or with a text editor." "","Lines beginning with # are comments." "","Lines beginning with > are

variables: [VAR_NAME], [VAR_VALUE]" ""," Lines beginning with $ are threads: [THREAD_NUMBER], [RED], [GREEN], [BLUE], [DESCRIPTION], [CATALOG_NUMBER]" ""," Lines beginning with * are stitch entries: [STITCH_TYPE], [X], [Y]" "," " ""," Stitch Entry Notes:" "," STITCH instructs the machine to move to the position [X][Y] and then make a stitch." ""," JUMP instructs the machine to move to the position [X][Y] without making a stitch." ""," TRIM instructs the machine to cut the thread before moving to the position [X][Y] without making a stitch." ""," COLOR instructs the machine to stop temporarily so that the user can change to a different color thread before resuming." ""," END instructs the machine that the design is completed and there are no further instructions." ""," UNKNOWN encompasses instructions that may not be supported currently." "," [X] and [Y] are absolute coordinates in millimeters (mm)." "#"," "

thread_color Sigma_Polyester_codes[] = { {"White", 0xFFFFFFFF, 10}, {"Black", 0xFF000000, 20}, {"Light Neon Green", 0xFFEDFF50, 21}, {"Neon Green", 0xFF96E845, 32}, {"Light Neon Orange", 0xFFFFE756, 33}, {"Med Neon Orange", 0xFFFF7824, 43}, {"Neon Pink", 0xFFF28DA6, 46}, {"Neon Orange Pink", 0xFFC70C57, 47}, {"Silver", 0xFFE22D2A, 101}, {"Silver Diamond", 0xFFB8B8B8, 102}, {"Lava Stone", 0xFF889186, 112}, {"Medium Grey", 0xFF737F7F, 115}, {"Dark Platinum", 0xFF565E5A, 116}, {"Charcoal", 0xFF515250, 117}, {"Badger Grey", 0xFF787668, 118}, {"Pumpkin Orange", 0xFFED572F, 135}, {"Turquoise", 0xFF2EA59C, 138}, {"Dark Wedgewood", 0xFF396276, 142}, {"Cardinal Red", 0xFF9B3B40, 213}, {"Maroon", 0xFF6C3E47, 216}, {"Rust", 0xFFBA6E4D, 253}, {"Medium Rust", 0xFFBB3D2E, 255}, {"Natural Pink", 0xFFF9DFCF, 301}, {"Baby Pink", 0xFFFBDED6, 303}, {"Piggy Pink", 0xFFF7CDD5, 304}, {"Sweet Pink", 0xFFF2AFB4, 305}, {"Blushing Pink", 0xFFE8418C, 307}, {"Pink", 0xFFE77F9D, 309}, {"Rose Pink", 0xFFF06F8C, 313}, {"Green", 0xFF008340, 317}, {"Shocking Pink", 0xFFDF99B6, 321}, {"Ruby", 0xFF820052, 325}, {"Garnet", 0xFFB1415F, 333}, {"Light Purple", 0xFFC394AE, 345}, {"Medium Purple", 0xFFA86E91, 347}, {"Dark Grape", 0xFF694169, 348}, {"Pastel Light Pink", 0xFFE6CFD5, 376}, {"Light Baby Blue", 0xFFA8BED7, 379}, {"Crystal Blue", 0xFFA0BFD7, 380}, {"Very Light Lavender", 0xFF90A6C6, 381}, {"Cornflower", 0xFF8FAFC6, 382}, {"Lavender", 0xFFB1B8D3, 383}, {"Denim", 0xFF416C9B, 385}, {"Light Violet", 0xFF7D77AF, 386}, {"Misty Rose", 0xFFFADAF4, 387}, {"Grape", 0xFF664090, 390}, {"Lt. Weathered Blue", 0xFFEAF0F9, 402}, {"Baby Blue", 0xFFA6D8F6, 403}, {"Med Baby Blue", 0xFF7B9CB0, 404}, {"Med Pastel Blue", 0xFF648DC7, 406}, {"Blue Raspberry", 0xFF3D6AA1, 409}, {"Med Royal Blue", 0xFF2D4491, 413}, {"Ocean Blue", 0xFF143D7A, 414}, {"Med Navy", 0xFF113263, 415}, {"Dark Navy", 0xFF0E1F38, 423}, {"Bright Sunshine", 0xFF0E1F38, 432}, /* TODO: duplicate case value / **{"Teal", 0xFF0091A5, 443}, {"Deep Teal", 0xFF005B63, 448}, {"Dark Teal", 0xFF00474D, 449}, {"Old Gold", 0xFFE5B15C, 466}, {"Cream", 0xFFD5BF9B, 501}, {"Pale Salmon", 0xFFFFD085, 503}, {"Med Peach", 0xFFF6B08E, 505}, {"Pink Salmon", 0xFFB3E851, 506}, {"Dark Peach", 0xFFF1A236, 508}, {"Dark Brown", 0xFF6E4337, 513}, {"Pale Red", 0xFFD8493E, 527}, {"Heron Blue", 0xFF697698, 541}, {"Pale Yellow", 0xFFFDE896, 601}, {"Pastel Yellow", 0xFFEDE55D, 602}, {"Golden Puppy", 0xFFDFA200, 609}, {"Buttercup", 0xFFFDE896, 612},** / TODO: duplicate case value / **{"Treasure Gold", 0xFFCEB24C, 616}, {"Old Gold", 0xFFAD953E, 619}, {"Pale Apricot", 0xFFFEF9EA, 627}, {"Tan", 0xFFBD9565, 628}, {"Mellow Yellow", 0xFFFDF76C, 632}, {"Lemon", 0xFFEDEF05, 633}, {"Amber", 0xFFF8C300, 646}, {"Mandarina", 0xFFE77817, 649}, {"Orange", 0xFFE66535, 650}, {"Golden Rod", 0xFFC69632, 652}, {"Light Olive", 0xFF98996D, 653}, {"Bright Gold", 0xFFC98300, 654}, {"Blue-Green", 0xFF007B8D, 688}, {"Forrest Green", 0xFF004D3D, 695}, {"Midnight Blue", 0xFF007EBA, 697}, {"Med Red", 0xFFCF0040, 700}, {"Med Blue", 0xFF28438C, 809}, {"Sweet Apricot", 0xFFD0B478, 812}, {"Skin", 0xFFE5BE6C, 818}, {"Jade", 0xFF449284, 825}, {"Light Silver", 0xFFCFCFCF, 829}, {"Papaya Whip", 0xFFDC875E, 831}, {"Cooper", 0xFFB4705D, 832}, {"Light Pecan", 0xFF9B5C4B, 833}, {"Burnt Rust", 0xFFA93121, 838}, {"Vegas Gold", 0xFFB18B00, 842}, {"Med Brown", 0xFF86462E, 857}, {"Med**

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SVG Colors

- Converted from the table at:
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- NOTE: This supports both UK and US English names, so the repeated values
- aren't an error. ; thread_color svg_color_codes[] = { "aliceblue", 0xFFf0f8ff, 0}, "antiquewhite", 0xFFfaebd7, 1}, "aqua", 0xFF00ffff, 2}, "aquamarine", 0xFF7fffd4, 3}, "azure", 0xFFf0ffff, 4}, "beige", 0xFFf5f5dc, 5}, "bisque", 0xFFffe4c4, 6}, "black", 0xFF000000, 7}, "blanchedalmond", 0xFFffebcd, 8}, "blue", 0xFF0000ff, 9}, "blueviolet", 0xFF8a2be2, 10}, "brown", 0xFFa52a2a, 11}, "burlywood", 0xFFdeb81b, 12}, "cadetblue", 0xFF5f9ea0, 13}, "chartreuse", 0xFF7fff00, 14}, "chocolate", 0xFFd22d1e, 15}, "coral", 0xFFFF7f0e, 16}, "cornflowerblue", 0xFF1c5fed, 17}, "cornsilk", 0xFFfff8dc, 18}, "crimson", 0xFFdc0e3c, 19}, "cyan", 0xFF00ffff, 20}, "darkblue", 0xFF00008b, 21}, "darkcyan", 0xFF008b8b, 22}, "darkgoldenrod", 0xFFb81a0b, 23}, "darkgray", 0xFFa9a9a9, 24}, "darkgreen", 0xFF001c00, 25}, "darkgrey", 0xFFa9a9a9, 26}, "darkkhaki", 0xFFbdb76b, 27}, "darkmagenta", 0xFF8b008b, 28}, "darkolivegreen", 0xFF0d6b2f, 29}, "darkorange", 0xFFff8c00, 30}, "darkorchid", 0xFF3f0ecc, 31}, "darkred", 0xFF8b0000, 32}, "darksalmon", 0xFFe93c7a, 33}, "darkseagreen", 0xFF8fbc8f, 34}, "darkslateblue", 0xFF1e3d8b, 35}, "darkslategray", 0xFF2f4f4f, 36}, "darkslategrey", 0xFF2f4f4f, 37}, "darkturquoise", 0xFF00ced1, 38}, "darkviolet", 0xFF5e00d3, 39}, "deeppink", 0xFFff0e5d, 40}, "deepskyblue", 0xFF00bfff, 41}, "dimgray", 0xFF2d2d2d, 42}, "dimgrey", 0xFF2d2d2d, 43}, "dodgerblue", 0xFF1e5aff, 44}, "firebrick", 0xFFb20a0a, 45}, "floralwhite", 0xFFfffaf0, 46}, "forestgreen", 0xFF0a8b0a, 47}, "fuchsia", 0xFFff00ff, 48}, "gainsboro", 0xFFdcdcdc, 49}, "ghostwhite", 0xFFf8f8ff, 50}, "gold", 0xFFffd700, 51}, "goldenrod", 0xFFdaa50e, 52}, "gray", 0xFF0e0e0e, 53}, "grey", 0xFF0e0e0e, 54}, "green", 0xFF000e00, 55}, "greenyellow", 0xFFadff2f, 56}, "honeydew", 0xFFf0fff0, 57}, "hotpink", 0xFFff2db4, 58}, "indianred", 0xFFcd5c5c, 59}, "indigo", 0xFF4b000a, 60}, "ivory", 0xFFffffff0, 61}, "khaki", 0xFFf0e68c, 62}, "lavender", 0xFFe6e6fa, 63}, "lavenderblush", 0xFFfff0f5, 64}, "lawngreen", 0xFF7cfc00, 65}, "lemonchiffon", 0xFFfffacd, 66}, "lightblue", 0xFFadd8e6, 67}, "lightcoral", 0xFFf00e0e, 68}, "lightcyan", 0xFFe0ffff, 69}, "lightgoldenrodyellow", 0xFFfafad2, 70}, "lightgray", 0xFFd3d3d3, 71}, "lightgreen", 0xFF5aee5a, 72}, "lightgrey", 0xFFd3d3d3, 73}, "lightpink", 0xFFffb6c1, 74}, "lightsalmon", 0xFFffa07a, 75}, "lightseagreen", 0xFF0eb2aa, 76}, "lightskyblue", 0xFF1bcefa, 77}, "lightslategray", 0xFF4d3a3f, 78}, "lightslategrey", 0xFF4d3a3f, 79}, "lightsteelblue", 0xFFb0c4de, 80}, "lightyellow", 0xFFffffe0, 81}, "lime", 0xFF00ff00, 82}, "limegreen", 0xFF0ecd0e, 83}, "linen", 0xFFfaf0e6, 84}, "magenta", 0xFFff00ff, 85}, "maroon", 0xFF0e0000, 86}, "mediumaquamarine", 0xFF2acdaa, 87}, "mediumblue", 0xFF0000cd, 88}, "mediumorchid", 0xFFba0dd3, 89}, "mediumpurple", 0xFF5d2edb, 90}, "mediumseagreen", 0xFF3cb32f, 91}, "mediumslateblue", 0xFF7b2cee, 92}, "mediumspringgreen", 0xFF00fa9a, 93}, "mediumturquoise", 0xFF1ed1cc, 94}, "mediumvioletred", 0xFFc70f0d, 95}, "midnightblue", 0xFF0d0d2e, 96}, "mintcream", 0xFFf5fffa, 97}, "mistyrose", 0xFFffe4e1, 98}, "moccasin", 0xFFffe4b5, 99}, "navajowhite", 0xFFffdead, 100}, "navy", 0xFF00000e, 101}, "oldlace", 0xFFfdf5e6, 102}, "olive", 0xFF0e0e00, 103}, "olivedrab", 0xFF6b8e0b, 104}, "orange", 0xFFffa500, 105}, "orangered", 0xFFff2d00, 106}, "orchid",

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Z102 Isacord Polyester Codes

thread_color Z102_Isacord_Polyester_codes[] = { {"?", 0xFFF8FFFF, 17}, {"?", 0xFF000000, 20}, {"?", 0xFFB7BABA, 105}, {"?", 0xFF73787A, 108}, {"?", 0xFF454B58, 138}, {"?", 0xFF9EA9A6, 142}, {"?", 0xFFC8C6BD, 150}, {"?", 0xFFFAEE5C, 220}, {"?", 0xFFE5CB4F, 221}, {"?", 0xFFFFF46A, 230}, {"?", 0xFFFEF9D9, 270}, {"?", 0xFFFFDC00, 311}, {"?", 0xFF624F00, 345}, {"?", 0xFFB8B25A, 352}, {"?", 0xFF8D8F5B, 453}, {"?", 0xFFFFF4A5, 520}, {"?", 0xFFB98E03, 542}, {"?", 0xFFE4C180, 651}, {"?", 0xFFC5BFA6, 672}, {"?", 0xFF96836D, 722}, {"?", 0xFF4E3500, 747}, {"?", 0xFFDDCBA5, 761}, {"?", 0xFF605840, 776}, {"?", 0xFFFFAF02, 800}, {"?", 0xFFF6AE32, 811}, {"?", 0xFFC89334, 822}, {"?", 0xFFE59300, 824}, {"?", 0xFFC89340, 832}, {"?", 0xFF9E947F, 873}, {"?", 0xFFC8700B, 922}, {"?", 0xFFBB5704, 931}, {"?", 0xFFB19072, 1061}, {"?", 0xFFFF8101, 1102}, {"?", 0xFFB1500A, 1115}, {"?", 0xFFC09C72, 1123}, {"?", 0xFF843D07, 1134}, {"?", 0xFFD8A67D, 1141}, {"?", 0xFF82421B, 1154}, {"?", 0xFFFF7319, 1300}, {"?", 0xFFFF3D1E, 1305}, {"?", 0xFFBA4005, 1311}, {"?", 0xFFC73C13, 1312}, {"?", 0xFFE66B21, 1332}, {"?", 0xFF3D1C11, 1346}, {"?", 0xFFFFBC95, 1351}, {"?", 0xFFFFCC93, 1362}, {"?", 0xFF373732, 1375}, {"?", 0xFFFFAF94, 1532}, {"?", 0xFF5B4032, 1565}, {"?", 0xFFFF6046, 1600}, {"?", 0xFFFF6D71, 1753}, {"?", 0xFFEBBAAE, 1755}, {"?", 0xFFEB2D2B, 1805}, {"?", 0xFFFF988F, 1840}, {"?", 0xFF434331, 1874}, {"?", 0xFFC11914, 1902}, {"?", 0xFFC8100D, 1903}, {"?", 0xFFBF0A21, 1906}, {"?", 0xFFD23C3E, 1921}, {"?", 0xFF8F8C93, 1972}, {"?", 0xFFA31A1C, 2011}, {"?", 0xFF4D0308, 2115}, {"?", 0xFFFFCDCC, 2155}, {"?", 0xFF871C45, 2500}, {"?", 0xFFDB4083, 2508}, {"?", 0xFFFF668A, 2520}, {"?",

0xFFC91243, 2521}, {"?", 0xFFFFA0B6, 2530}, {"?", 0xFFFEA5B9, 2550}, {"?", 0xFF626C7E, 2674}, {"?", 0xFF5E1942, 2711}, {"?", 0xFF33001D, 2715}, {"?", 0xFFA57B8D, 2764}, {"?", 0xFF702A69, 2810}, {"?", 0xFFB385BC, 2830}, {"?", 0xFF240047, 2900}, {"?", 0xFF724593, 2910}, {"?", 0xFF634D86, 2920}, {"?", 0xFF000136, 3110}, {"?", 0xFF000021, 3355}, {"?", 0xFF054ABD, 3522}, {"?", 0xFF1C005D, 3541}, {"?", 0xFF001F71, 3544}, {"?", 0xFF002E5E, 3622}, {"?", 0xFF71AAD8, 3630}, {"?", 0xFF001748, 3644}, {"?", 0xFFC8DBE4, 3650}, {"?", 0xFF9FC7DF, 3730}, {"?", 0xFF082E4D, 3743}, {"?", 0xFF98B0BC, 3750}, {"?", 0xFF23679C, 3810}, {"?", 0xFF3D657E, 3842}, {"?", 0xFF00669F, 3901}, {"?", 0xFF47AEDD, 3910}, {"?", 0xFFBBDFEB, 3962}, {"?", 0xFFBABEB7, 3971}, {"?", 0xFF015D7E, 4032}, {"?", 0xFFD5DDDB, 4071}, {"?", 0xFF888D8E, 4073}, {"?", 0xFF007CA6, 4103}, {"?", 0xFF3EBBC8, 4111}, {"?", 0xFF005C79, 4116}, {"?", 0xFF80CCD8, 4240}, {"?", 0xFFACCEC7, 4250}, {"?", 0xFF006E74, 4410}, {"?", 0xFF002A29, 4515}, {"?", 0xFF38A4AE, 4620}, {"?", 0xFFAFD8CD, 5050}, {"?", 0xFF149B7B, 5210}, {"?", 0xFF7AC8AF, 5220}, {"?", 0xFF187166, 5233}, {"?", 0xFF004B23, 5374}, {"?", 0xFF006835, 5415}, {"?", 0xFF5C9C51, 5531}, {"?", 0xFF0E9543, 5613}, {"?", 0xFF5E7A17, 5833}, {"?", 0xFF225926, 5944}, {"?", 0xFFBCD633, 6011}, {"?", 0xFFBBCD91, 6051}, {"?", 0xFF978B3C, 6133}, {"END", 0, -1} };